

Automatic Image Analysis

Visual Object Models

I. Hough Transform

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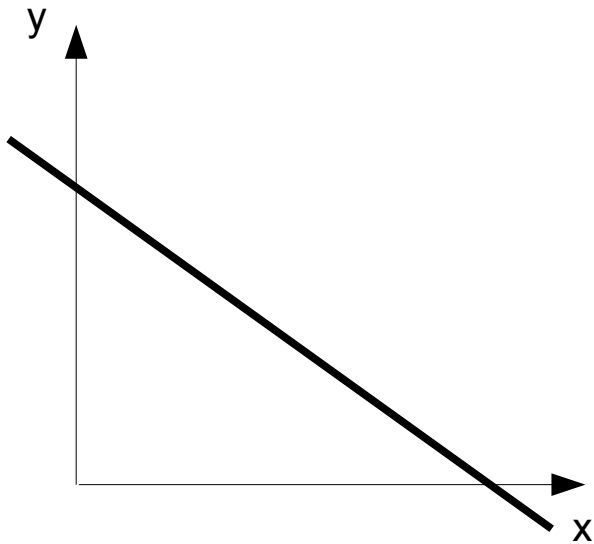
Object Model

- Concept of, how an object looks like.
- Give some examples!
- How do we express the object model computationally?

(ρ, θ) Invariance

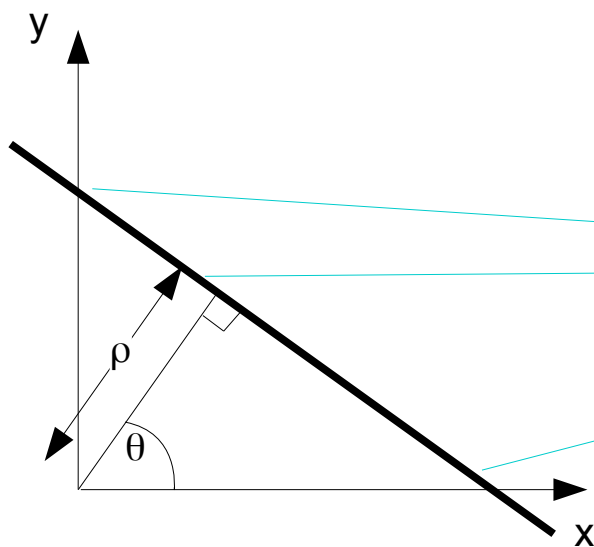
A simple example: Objects are straight lines

→ Might be disturbed by noise or interrupted.

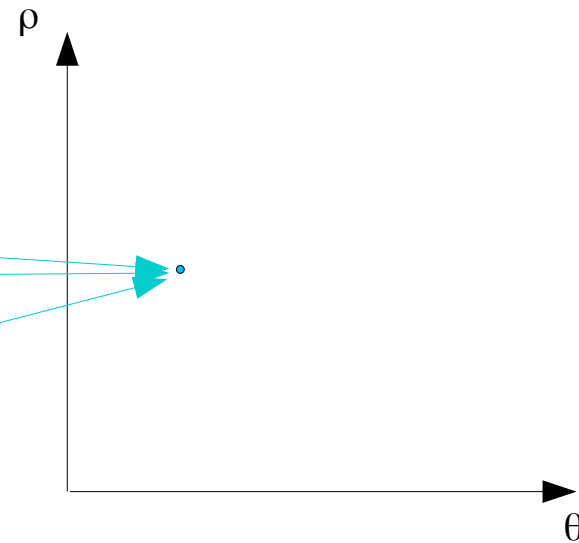


(ρ, θ) Invariance

A line in the image plane is defined by parameters (ρ, θ) :



In parameter space over (ρ, θ) , the line collapses to a point:



Invariant for all pixels on the line:

$$\begin{pmatrix} x \\ y \end{pmatrix}^T \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} = \rho$$

Line Detection

- 0) Compute (ρ, θ) at each line-pixel
- 1) Increment the parameter space at coordinate (ρ, θ)
- 2) A line leads to a pronounced maximum

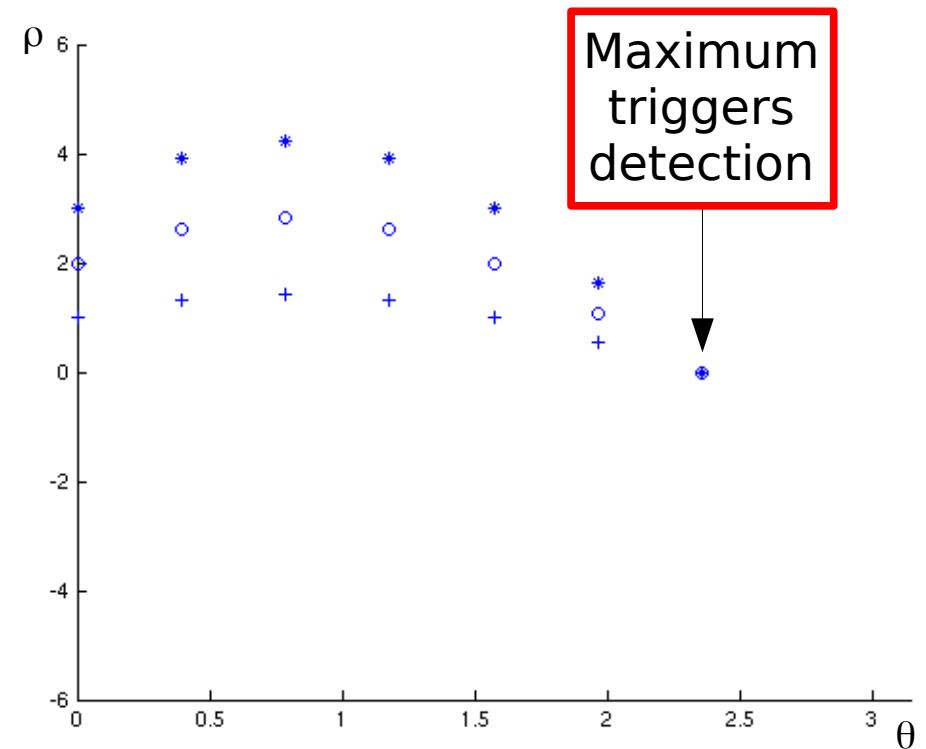
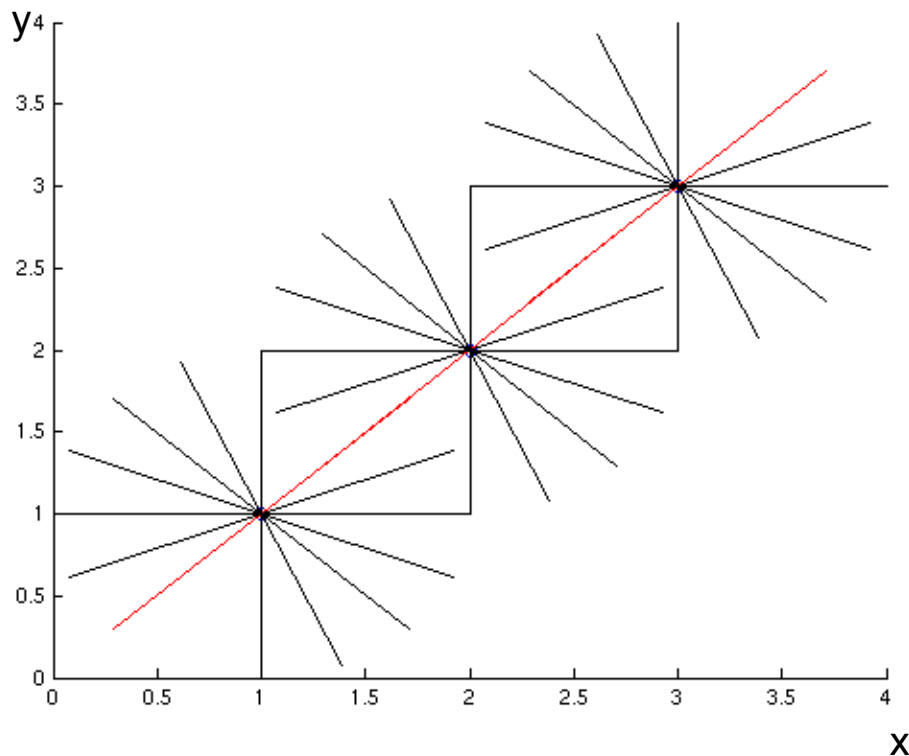
Radon Transformation

Problem: (ρ, θ) are unknowns

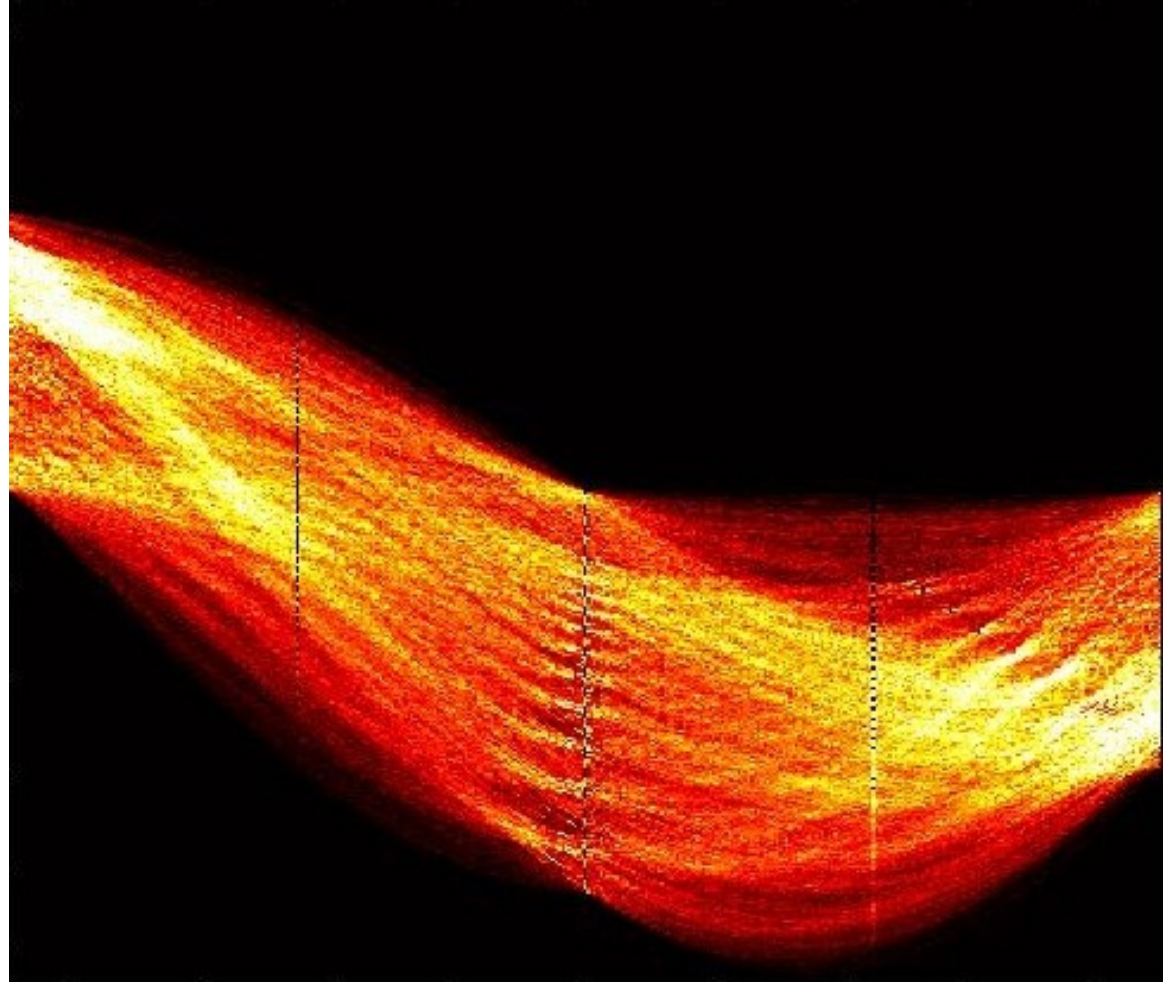
- Infinite number of potential lines intersect at every point

Solution:

- Each line through a point (x, y) is defined by fixing its direction θ .
- ρ can be computed from x , y and θ .
- For all directions θ : Compute ρ and increase element (ρ, θ) in parameter space



Radon Transformation



Hough Transformation

- Every edge has a position and a direction.
 - It can be part of only one (infinitely extended) line.
- Collinear edges point to a **single cell in parameter space**.

In case of points (rather than edges):

- A point is consistent with infinitely many lines.
- BUT: it is not consistent with all lines.

Points vote for every line they are consistent with.

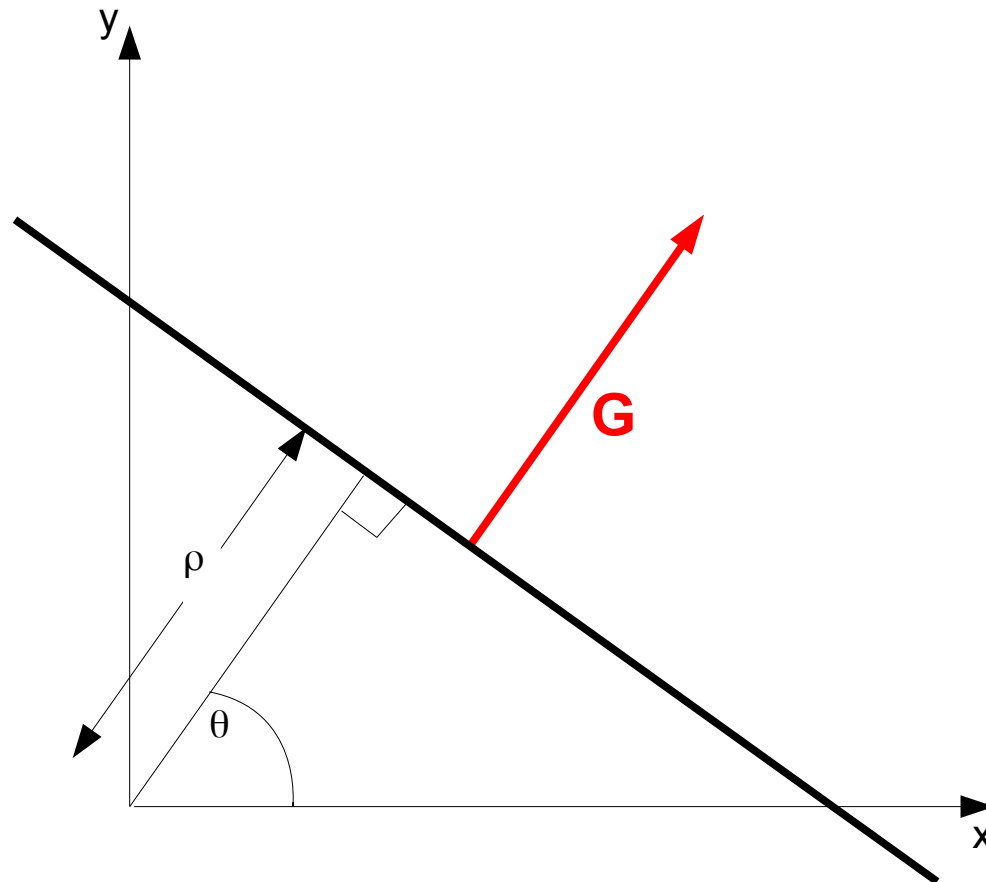
- Then it is more likely to find accidental mismatches.
- Higher threshold for peaks in parameter space are necessary.

BUT: Edge detection does not only provide

- edge **points**,
- but also edge **directions**!

Hough Straight Line Detector

- Alternative (more intelligent) solution: Use gradient information.
- The local gradient direction fixes θ , and (ρ, θ) can be calculated directly.
- It is no longer necessary to consider every possible line through each pixel!



Hough Line Detector

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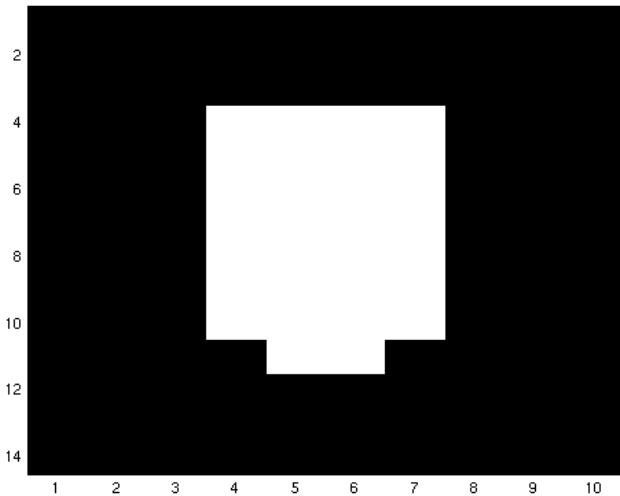
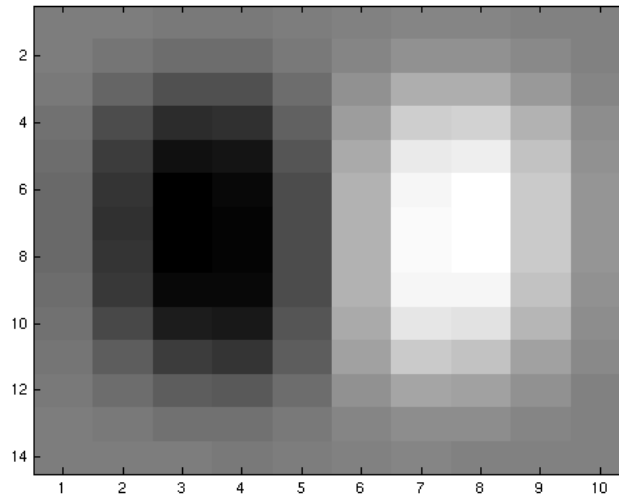
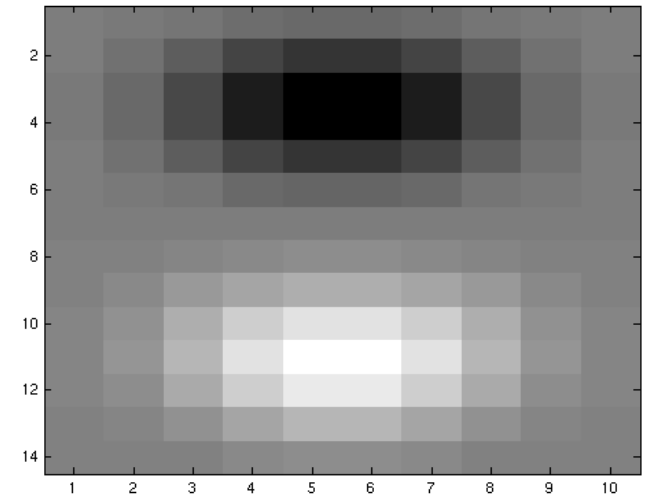


Image
 $B(x, y)$



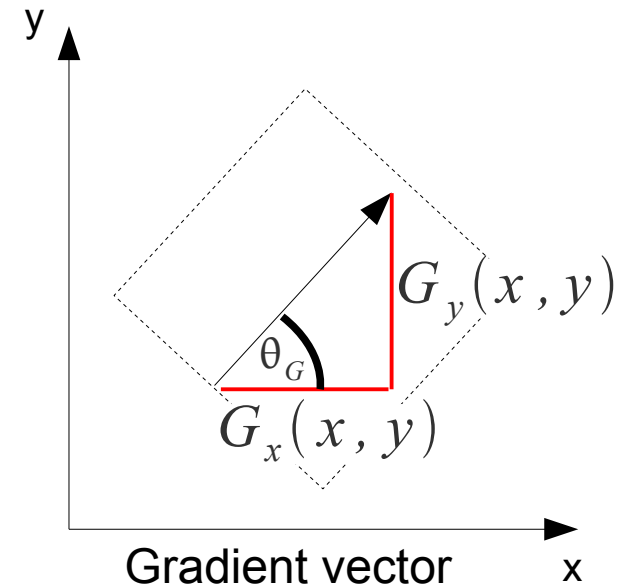
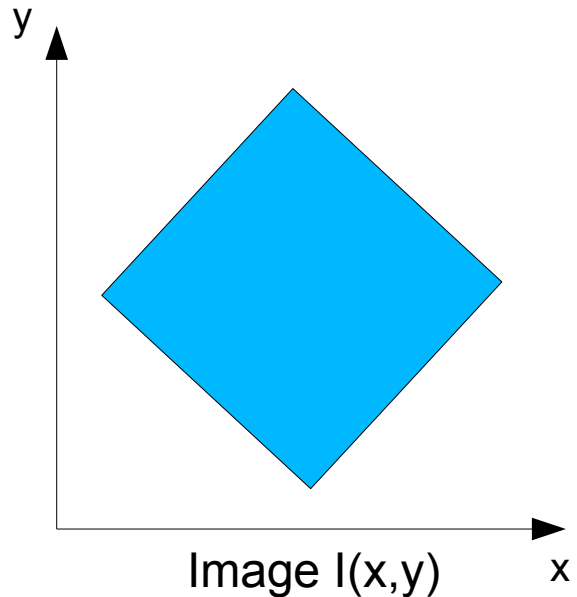
Gradient in x direction
 $G_x(x, y) = \frac{\partial}{\partial x} B(x, y)$



Gradient in y direction
 $G_y(x, y) = \frac{\partial}{\partial y} B(x, y)$

Hough Line Detector

- Each pixel (x,y) is associated with a gradient vector $(G_x(x,y), G_y(x,y))^T$.



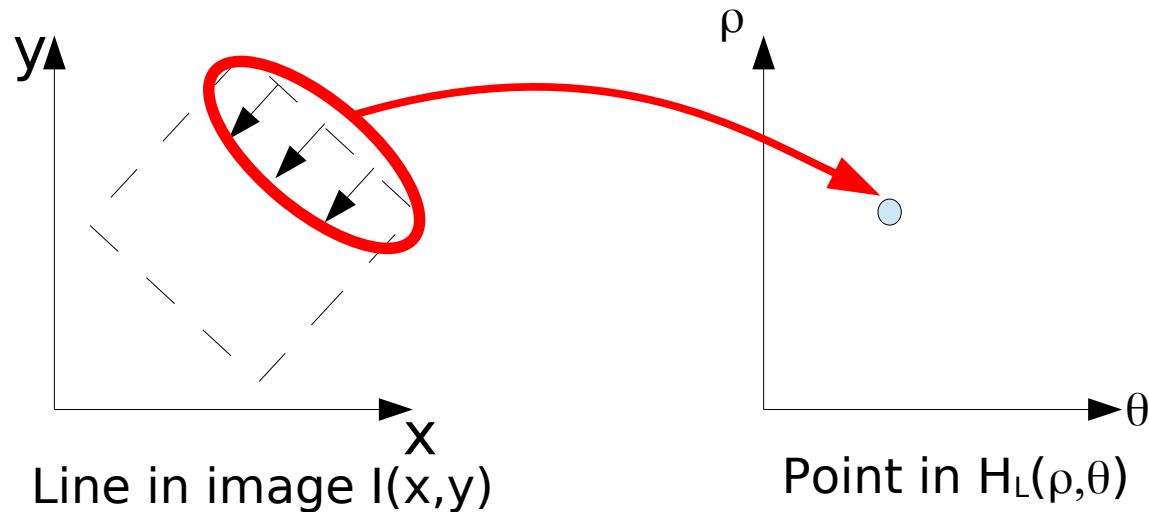
- The angle $\theta_G(x,y)$ is easy to obtain, e.g. in C++:

$$\theta_G(x,y) = \text{atan2}(G_y(x,y), G_x(x,y))$$

- Once $\theta_G(x,y)$ is known, coordinates (ρ, θ) in parameter space are uniquely determined:

$$\begin{pmatrix} x \\ y \end{pmatrix}^T \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} = \rho$$

Hough Line Detector

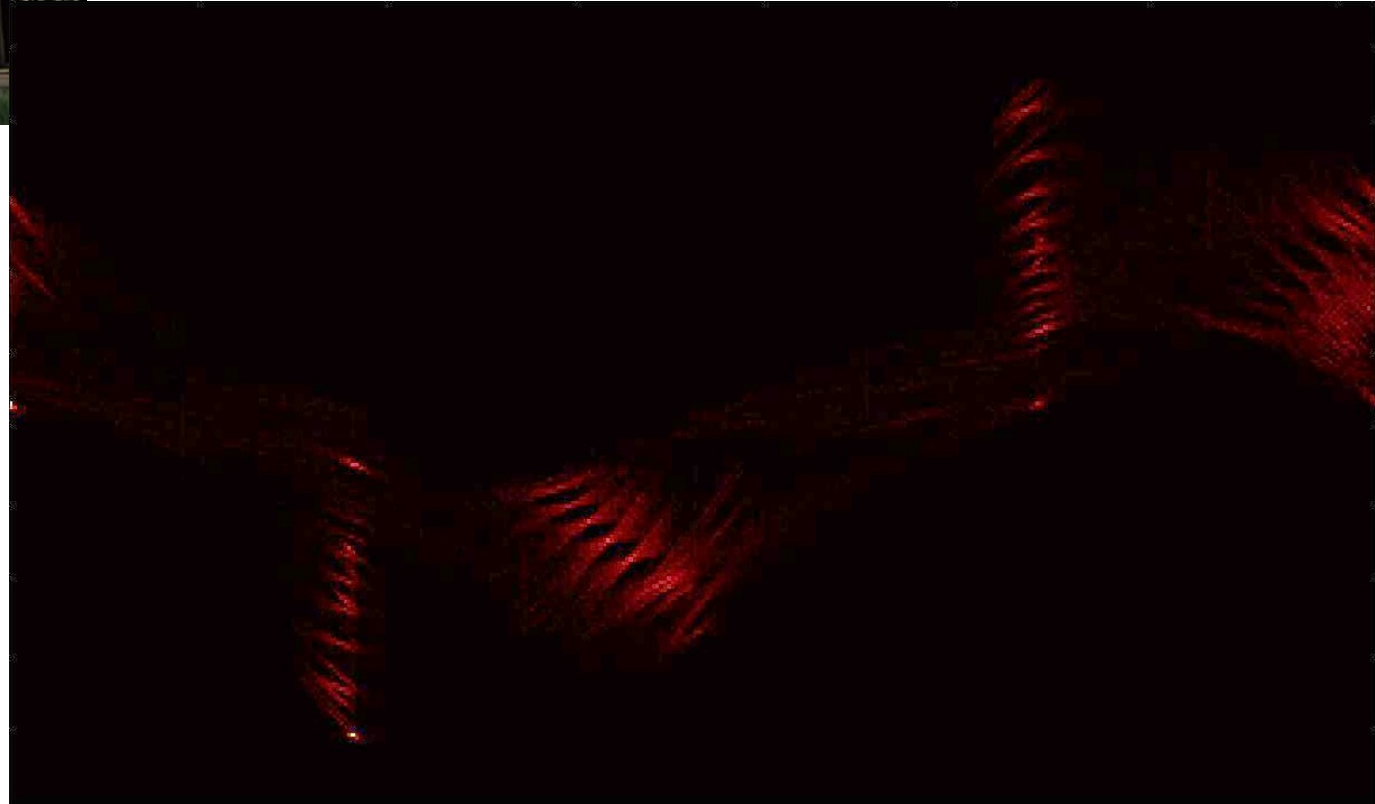


- 1) Compute gradients in x and y .
- 2) Determine ρ and θ using the local gradient direction.
- 3) Increase $H_L(\rho,\theta)$ by the local gradient magnitude:

$$H_L(\rho, \theta) \leftarrow H_L(\rho, \theta) + \sqrt{G_x(x, y)^2 + G_y(x, y)^2}$$

A line in $I(x,y)$ produces a pronounced maximum $H_L(\rho,\theta)$.

Hough Line Detector

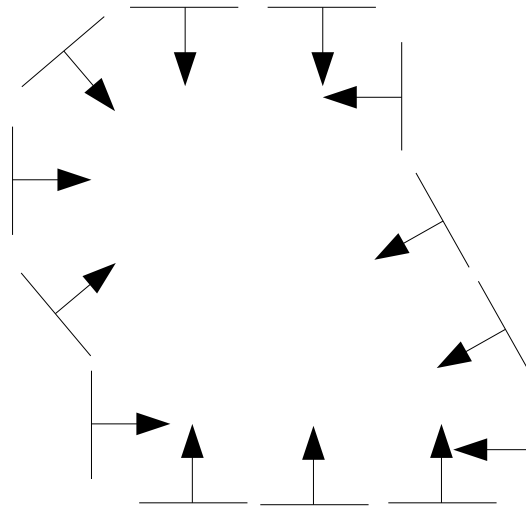


Visual Object Models

II. Generalized Hough Transform

Generalized Hough Transformation

Given: Edges extracted from an arbitrary 2D curve:



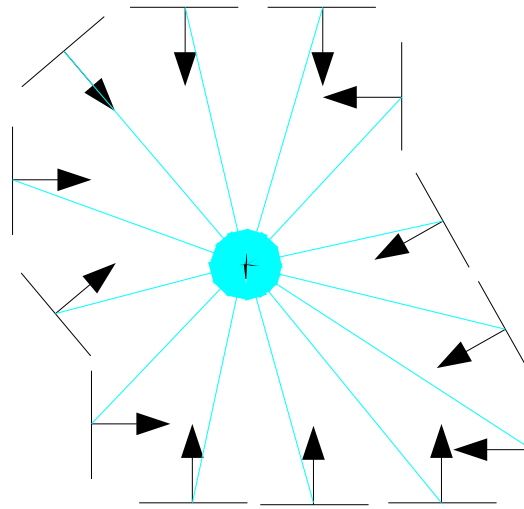
Can we use a Hough space to determine whether the curve is in another image?

If yes, under which transformations?

→ e.g. translation, scale, rotation, occlusion

Generalized Hough Transformation

Given: Edges extracted from an arbitrary 2D curve:



Select a reference point.

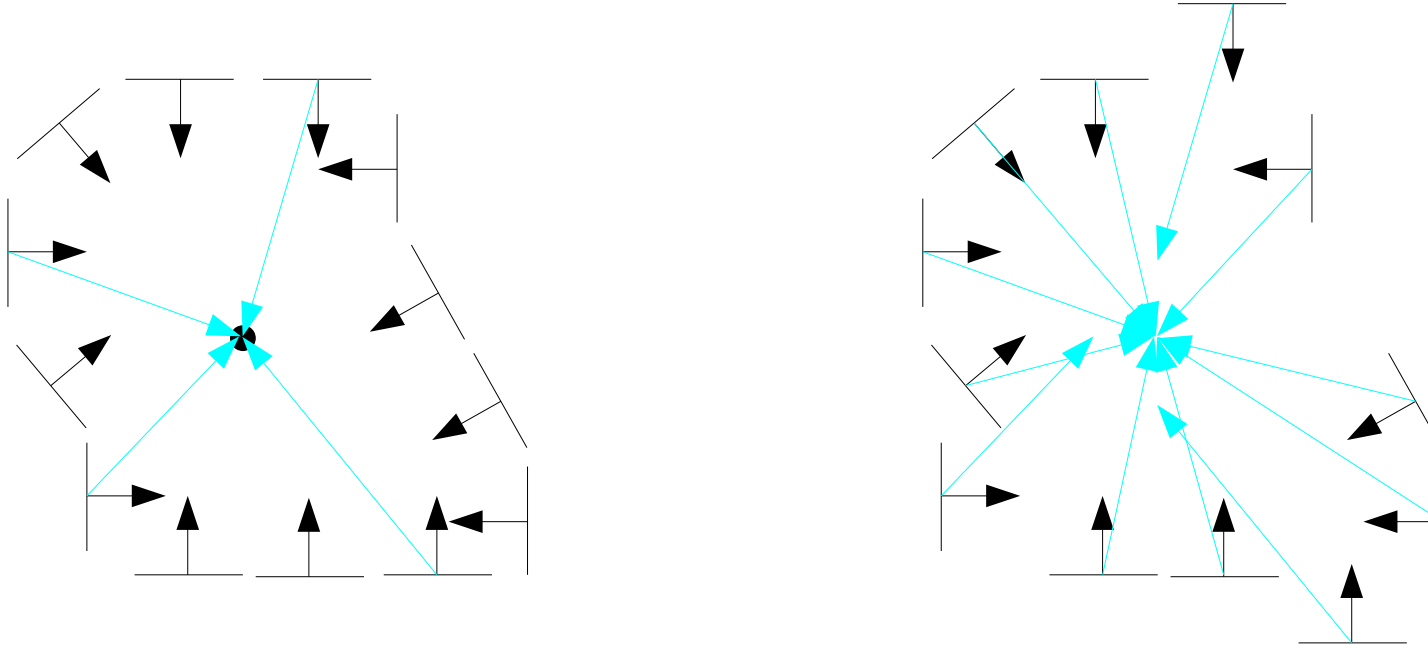
- Can be center of mass (but doesn't have to be).
- The Hough space parameters are the position of this point.

For every edge in the model curve,
there is a vector from the edge to the reference point.

- Store this set of vectors.

Generalized Hough Transformation

Given: Edges extracted from an arbitrary 2D curve:



Now consider the edges in your test data:

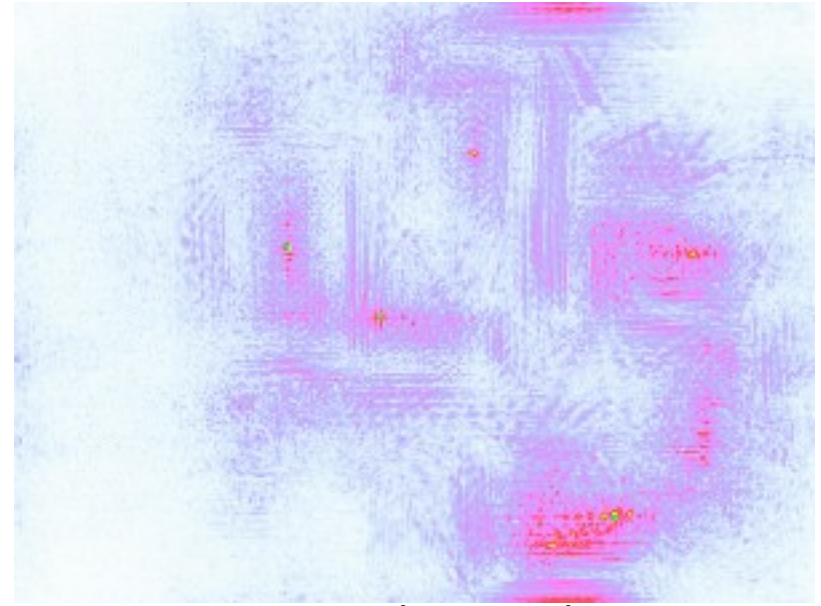
- If the edge is part of the curve, where could the reference point be?
- It must be one of the edges in the curve,
 - Offset from edge to reference point must be one of the stored vectors!
 - Vote often!
 - Peak in Hough space is the point with the most supporting edge points.

Generalized Hough Transformation

- An image $I(x,y)$ is processed to discover instances of an object.
- The object of interest is known from an example image $O(x,y)$.
- Processing is once again concerned with accumulating „votes“ in a parameter space $H_G(x,y,\theta,s)$.



Image $I(x,y)$



$H_G(x,y,\theta,s)$



Object $O(x,y)$

Pronounced maxima in $H_G(x,y,\theta,s)$ indicate that the image contains an object of interest centered at (x,y) , scaled by factor s , and rotated by angle θ .

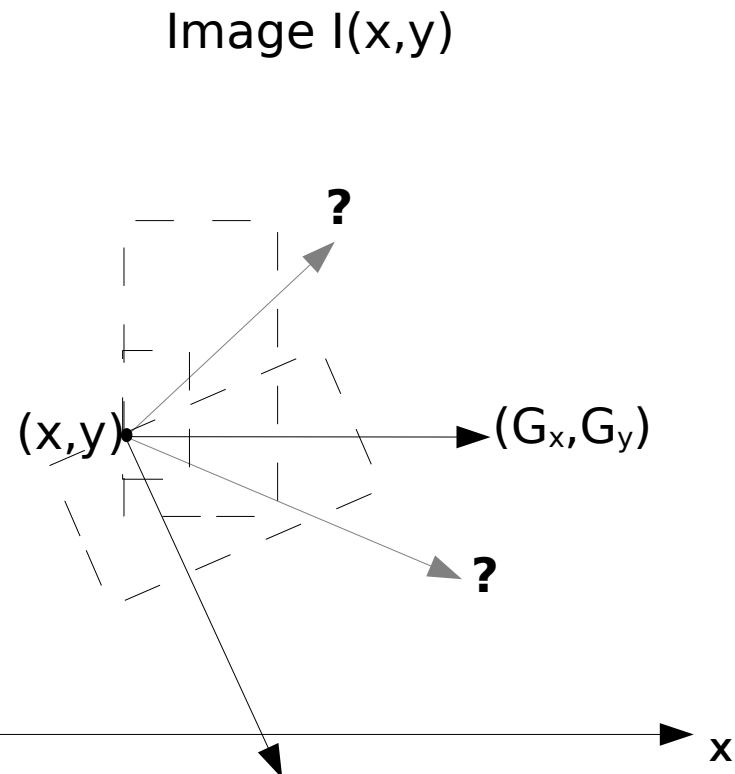
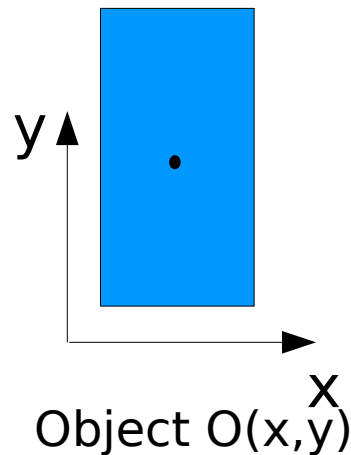
Generalized Hough Transformation

Given: the gradient vector (G_x, G_y) at each image location

Required: possible locations of the object center

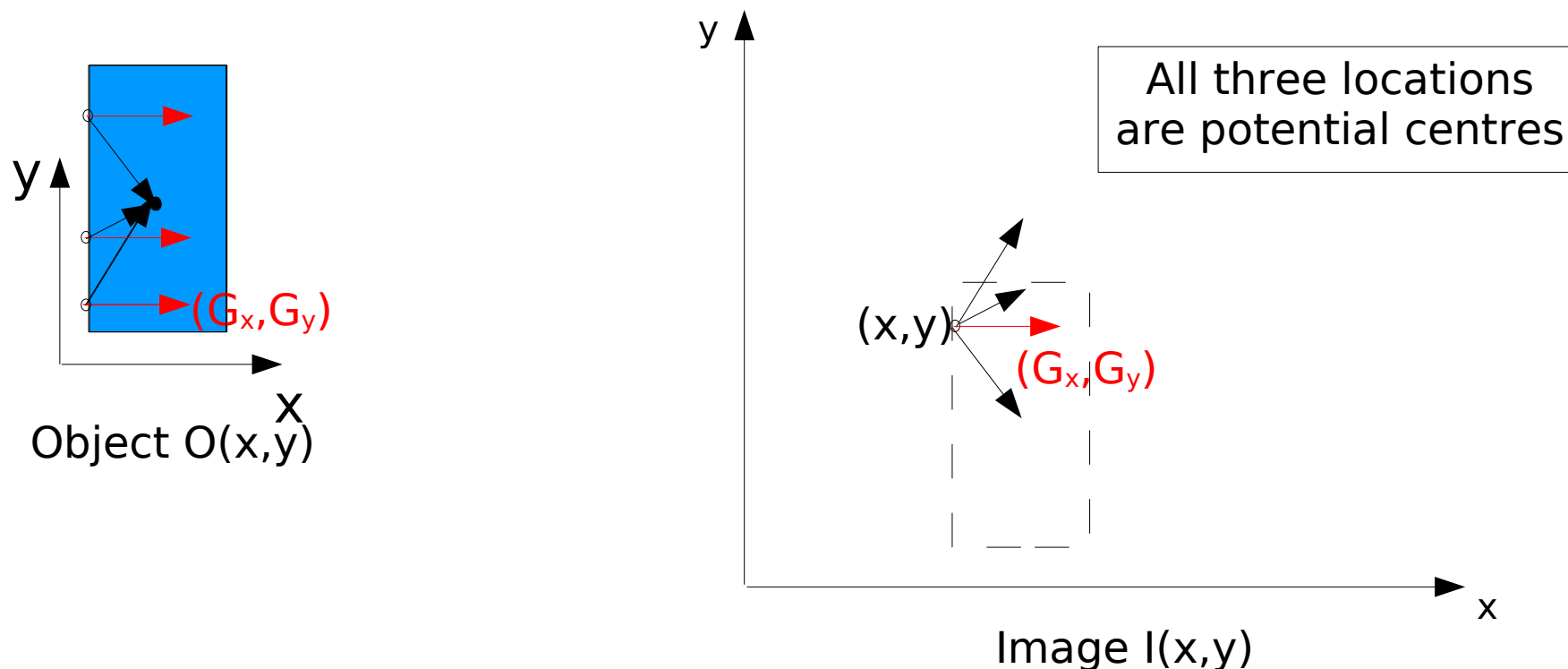
How can possible center locations be inferred from local gradient directions?

Which locations in $H_G(x, y, \theta, s)$ are to be increased (i.e. voted for)?



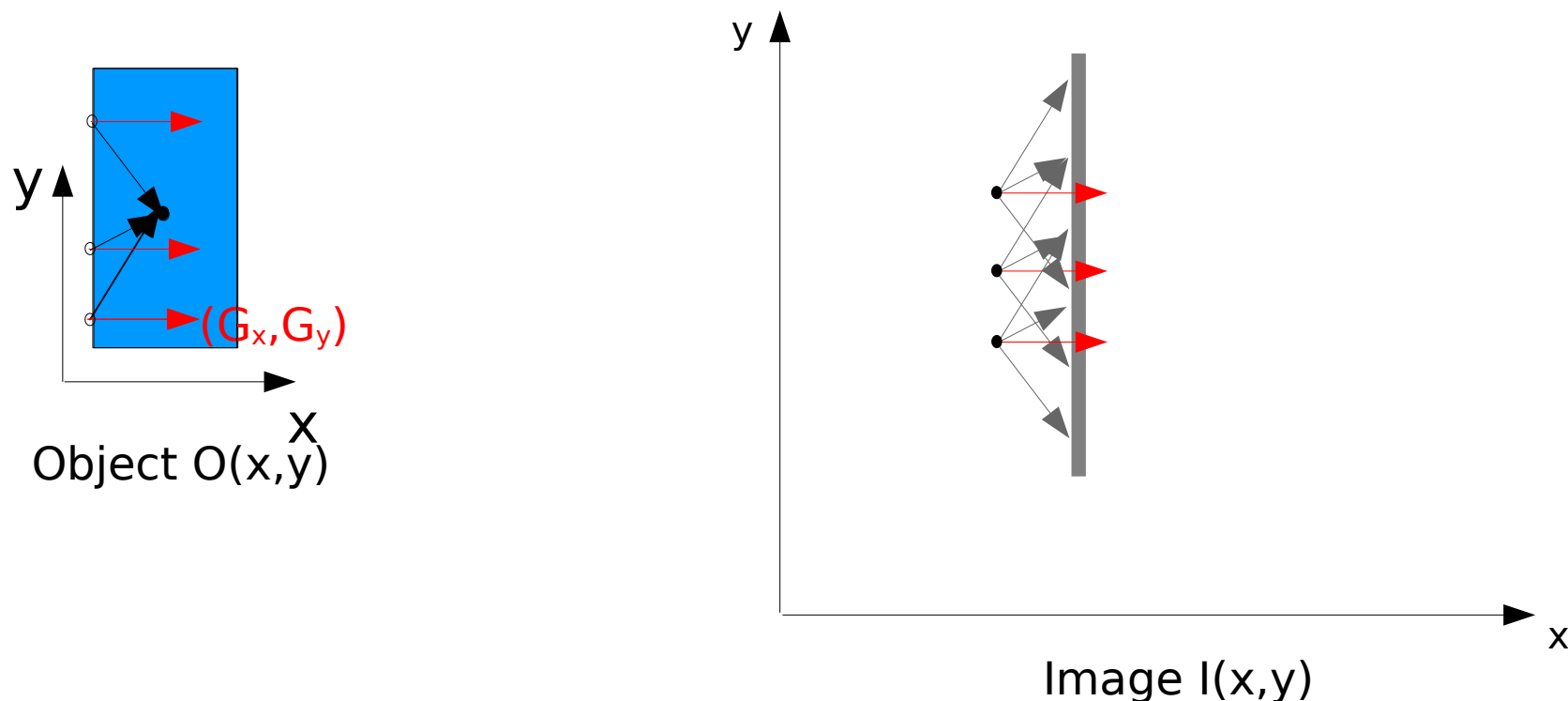
Generalized Hough Transformation

- Exploit relation between local gradient and object centre
- The local gradient is ambiguous:
 - The same gradient occurs at different points on the object.
 - A given gradient direction may correspond to several possible object centres.
 - In general, several locations in $H_G(x,y,\theta,s)$ must be increased.



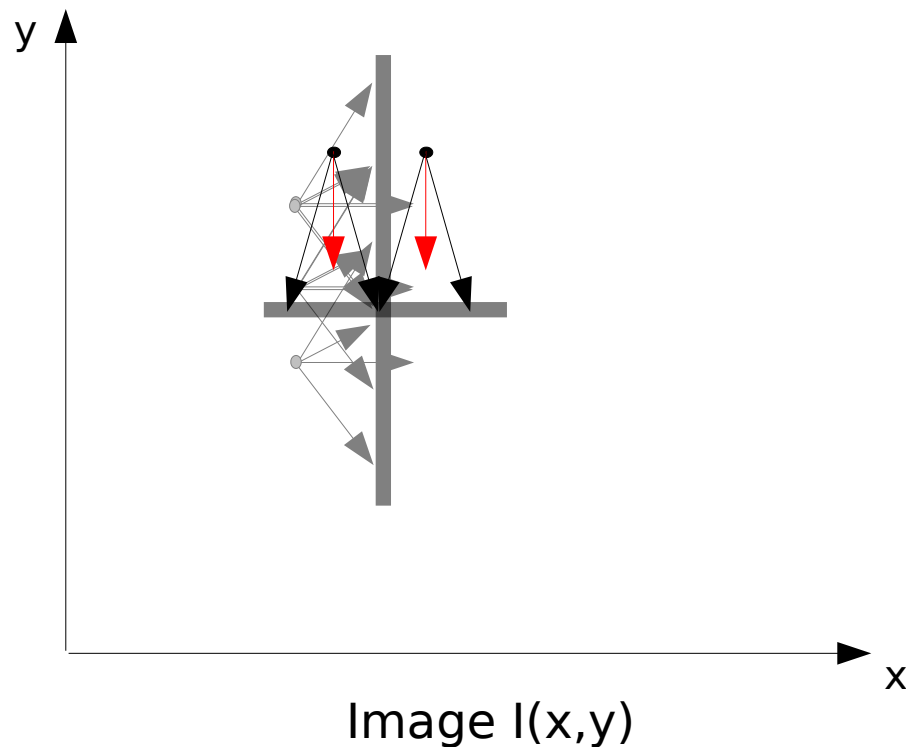
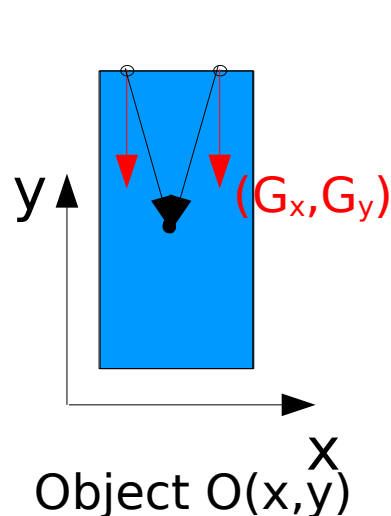
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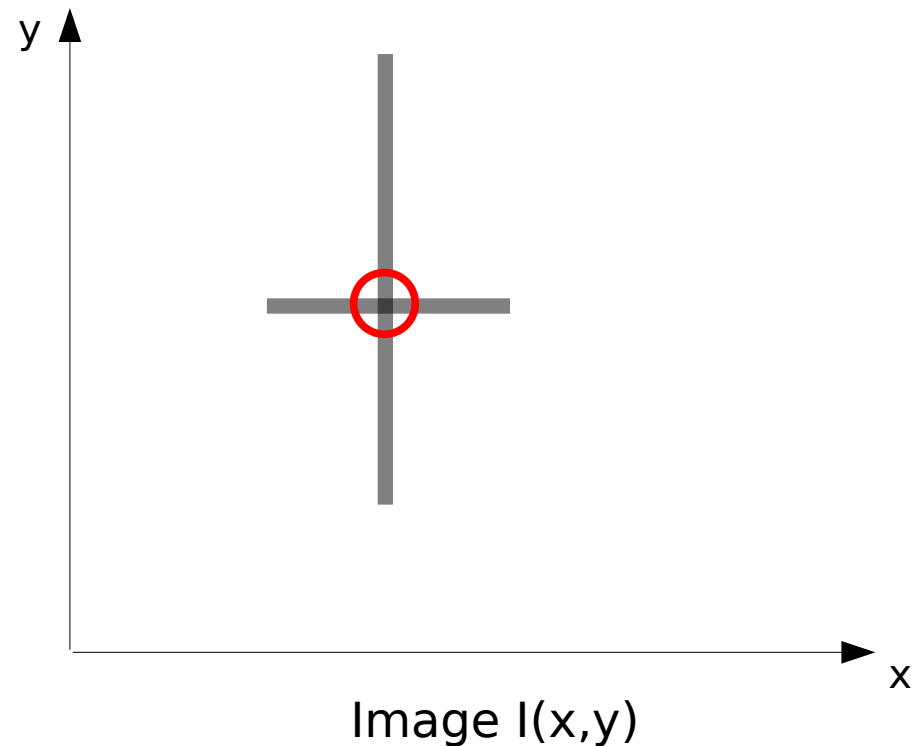
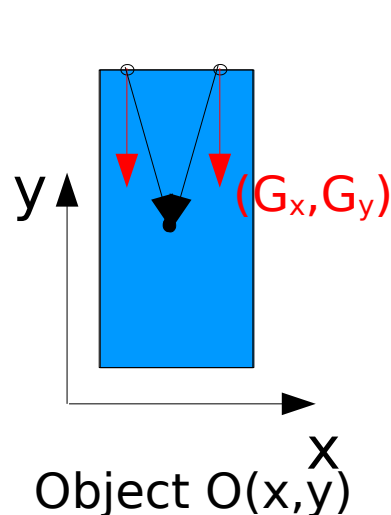
Generalized Hough Transformation

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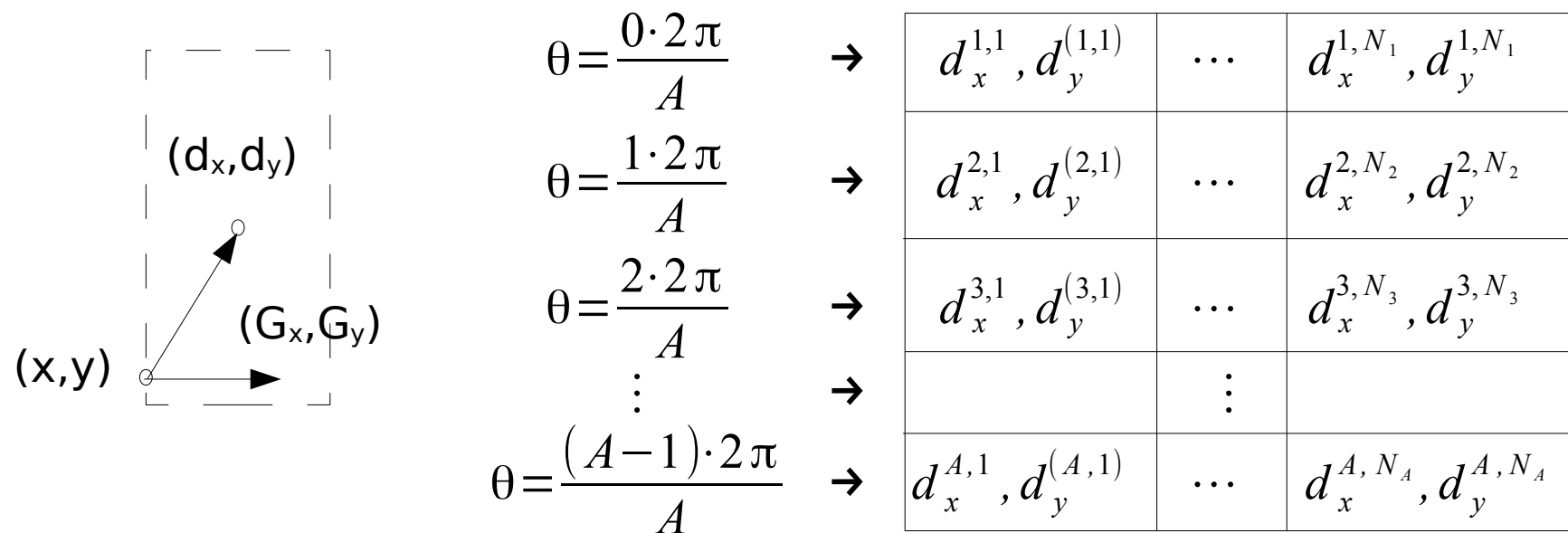
Generalized Hough Transformation

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Generating the R-Table

- The R-Table stores, for each pixel on the edge of an object O, the local gradient direction and the displacement from the pixel to the object center.



Row T of the R-Table

- Applies, when local gradient direction is close to $2\pi(T-1)/A$.
- The displacement to the centre is $(d_x^{T,1}, d_y^{T,1})$ or $(d_x^{T,2}, d_y^{T,2})$ or ... or $(d_x^{T,NT}, d_y^{T,NT})$.

Generalized Hough Transformation

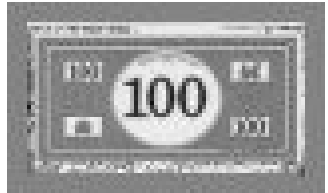
Training phase

1. Extract edge pixels from the template.
2. Store displacement vectors from edge position to reference point based on gradient direction in R-table.

Prediction phase

1. Extract edge pixels from the template.
2. for each scale s between minimal and maximal scale
Scale displacement vectors in R-table.
for each orientation θ between 0 and 180°
Rotate displacement vectors according to θ .
Circ-shift rows of R-table.
for each edge pixel
Identify corresponding row of R-table based on edge orientation.
Use all displacement vectors in this row to cast votes.
Increase $H(x,y,s,\theta)$ accordingly.
3. Find peaks in 4-dimensional Hough space $H(x,y,s,\theta)$.

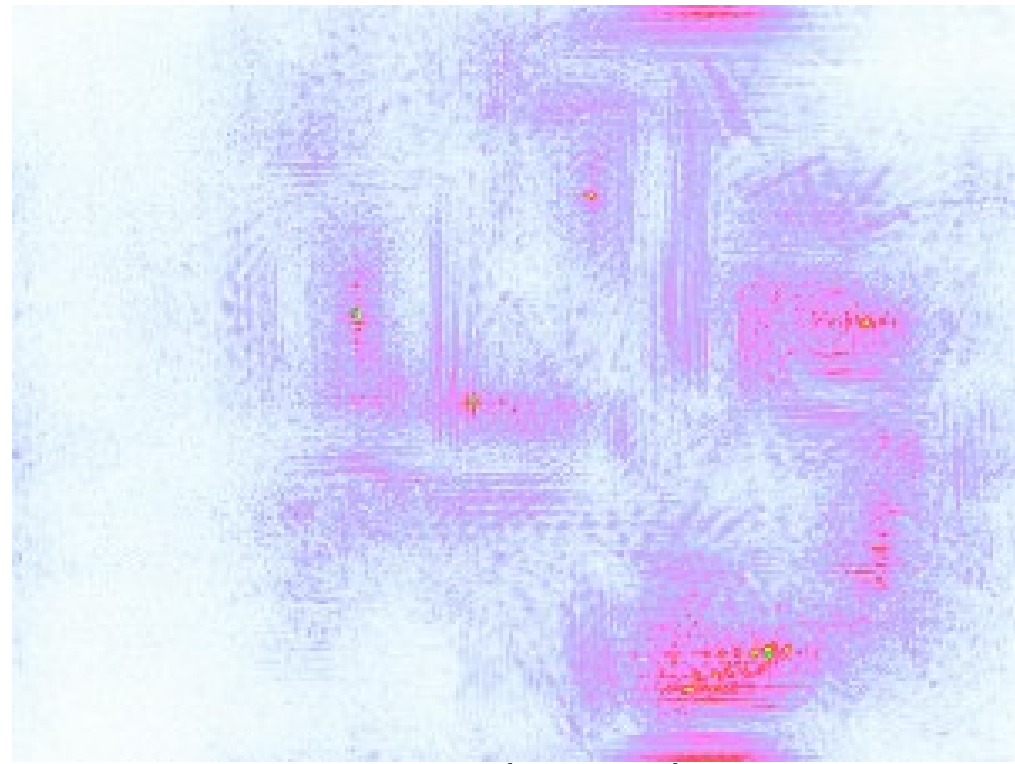
Generalised Hough Transformation



Object $O(x,y)$



Image $I(x,y)$



$H_G(x,y,\theta,s)$

Visual Object Models

III. Salient Points

Identifying Salient Points in Images

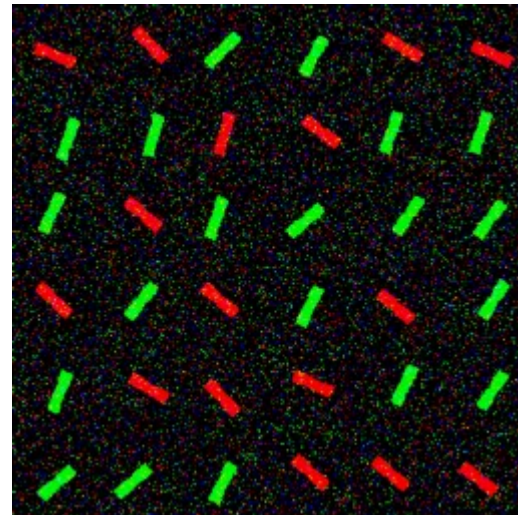
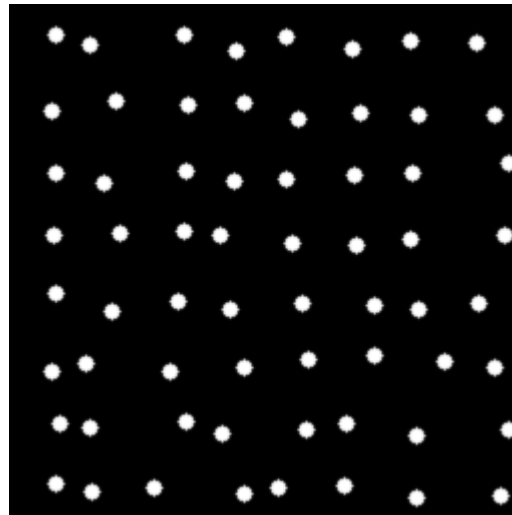
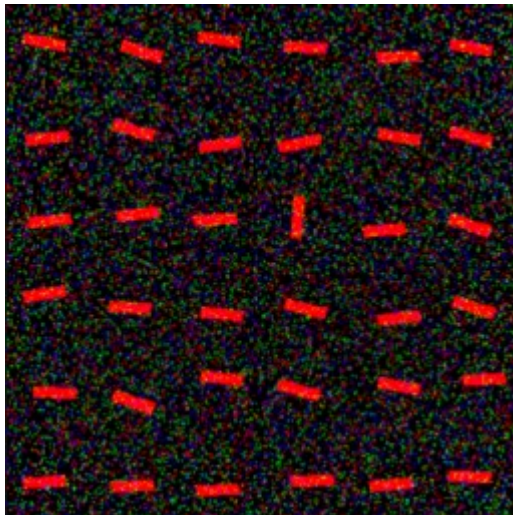
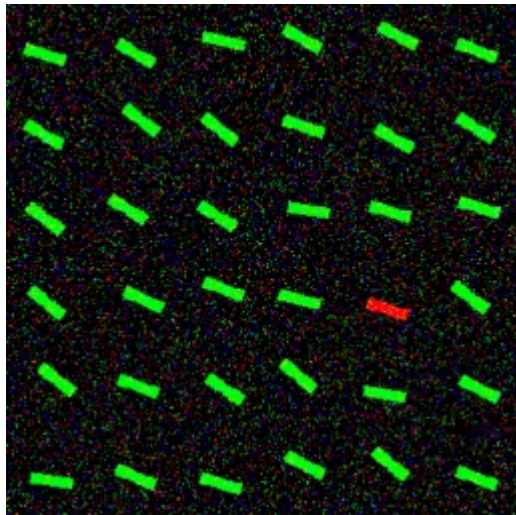
Biological Saliency



Saliency is a subjective perceptual quality of items which get prioritized attention by an observer.

Saliency

- Detection of (image) locations where local visual attributes significantly differ from the surrounding with respect to one property or a combination of properties
- Based on simple visual features of the early stages of cortical processing
color, edge orientation, luminance, or motion direction



A Computational Model of Saliency

- Analysis of visual information by early visual neurons
 - Sensitive to simple visual features
 - **Parallel** over the entire visual field
 - At **multiple spatial** and **temporal scales**
- Result: Several cortical feature maps
 - **Strength of visual feature** at every location **within visual field**
- Emphasize locations that differ from neighbours
- Combine all such locations in a single map
- **Attention** is focused on most salient locations
 - Triggers motor actions
(**adjust eyes and head**)

Salient Image Regions

Goal

- Detection of **salient / distinctive** regions in images
- Extraction of **significant** description of the image content
- Regions ideally correspond to object parts
- Scale and rotation invariant detection

But central problem

- What is “significant”?

One option:

- Geometric characteristics: interest-point operators
- see Module “Digital Image Processing”

Here: Information-theoretically more complex concept

- **Saliency as “Rareness”**
- T. Kadir, A. Zisserman, and M. Brady,
“An Affine Invariant Salient Region Detector”, ECCV 2004

Salient Image Regions

Saliency as “Rareness”:

- If high-grade differencing descriptors are used everything is rare.
- If descriptors are very general nothing is rare.
- Mathematical comprehensible saliency as local complexity
→ Shannon Entropy of local attributes

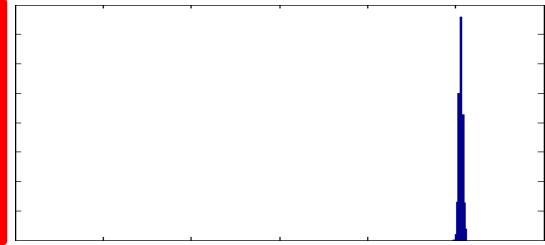
Entropy corresponds to

- A measure of uncertainty or randomness
- The average information content of a signal
- The minimal amount of bits necessary for encoding

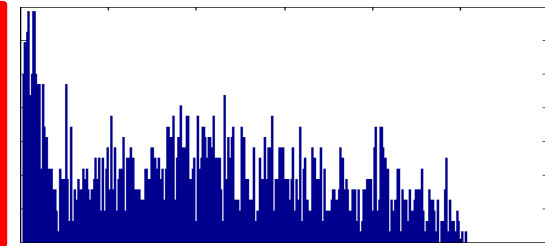
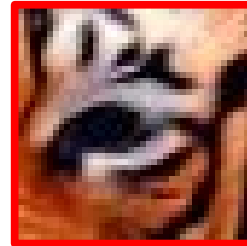
$$H = - \sum_{k=0}^{255} p(k) \cdot \log_2 p(k) \quad \text{with } 0 \cdot \log_2(0) = 0$$

Salient Image Regions

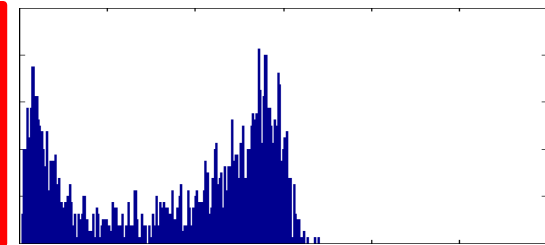
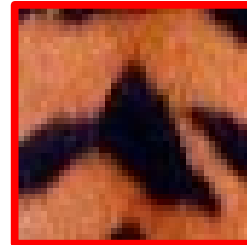
Local histograms of image intensity



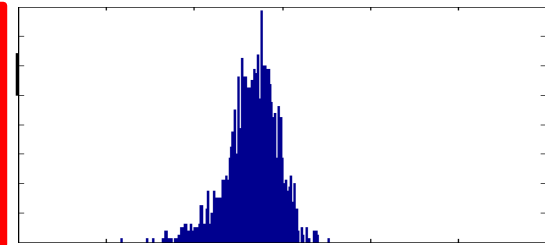
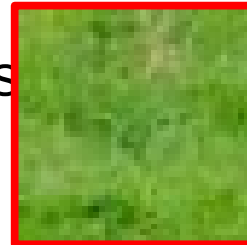
H=2.2



H=7.7



H=6.4



H=5.8

res

Salient Image Regions

Image Pyramids

- Create a multi-scale representation by explicitly varying the scale
- Sub-sample the image with logarithmic steps in resolution

$$\{R \tau^0, R \tau^{-1}, R \tau^{-2}, \dots, R \tau^{-N}\}$$

- Gather information at multiple scales

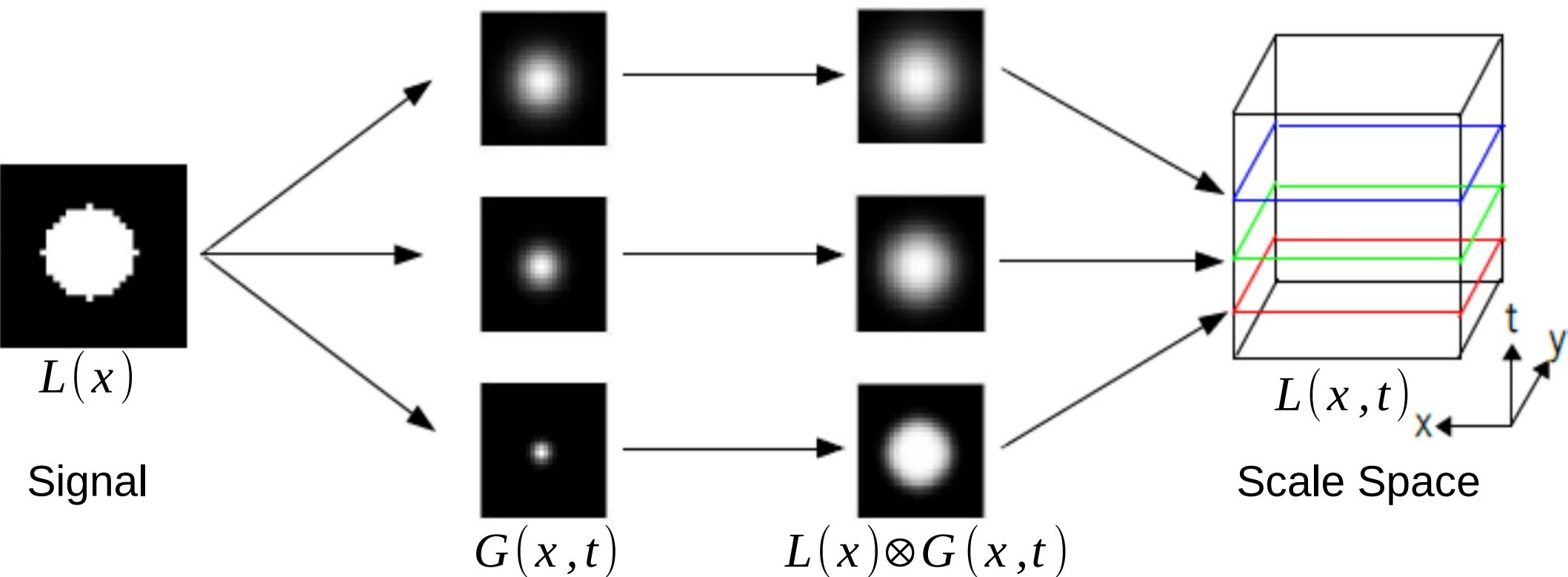


Salient Image Regions

Linear Gaussian Scale Space

- Initial signal: $L(x), x \in \mathbb{R}^n, x = \{x_1, x_2, \dots, x_N\}$
- Scale space:

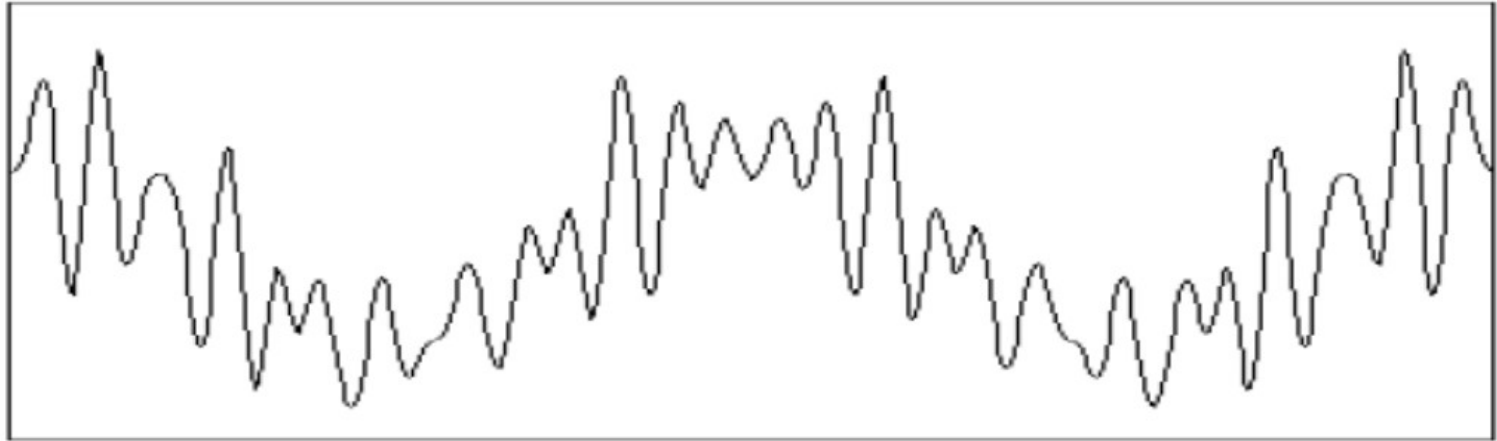
$$L(x, t) = L(x) \otimes G(x, t) \text{ where } G(x, t) = (\pi t)^{-n/2} \exp\left(-\frac{|x|^2}{t}\right)$$



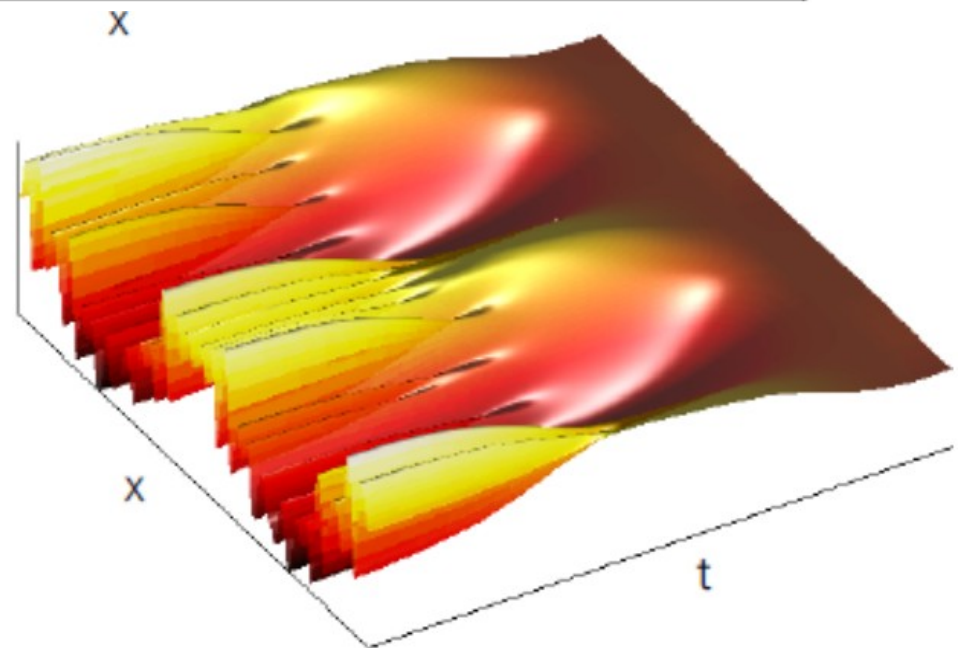
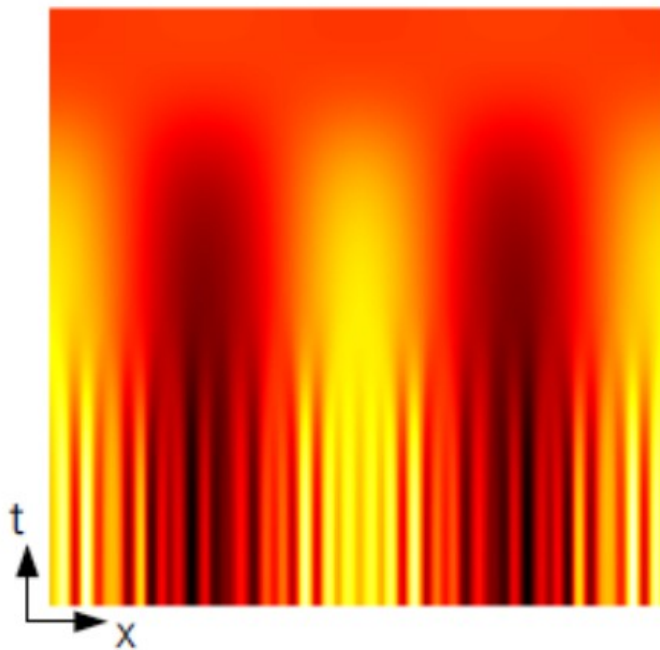
Salient Image Regions

Linear 1D Gaussian Scale Space

$L(x)$



$L(x, t)$



Salient Image Regions

Linear 2D Gaussian Scale Space



Salient Image Regions

Scale selection

- Saliency should be valid in spatial domain **and** in scale space.
- Self-similarity is not distinctive
- Neither in spatial domain nor in scale

Wanted:

- Measure of the “Self-Unsimilarity” in scale space
- Select scales at which entropy (as a function of scale) has peaks.

$$y(s_p, x) = H_D(s_p, x) W_D(s_p, x)$$

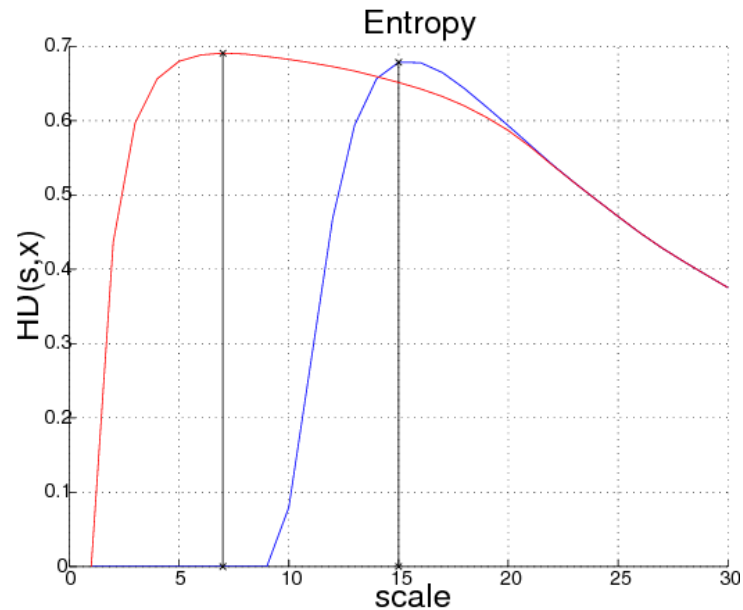
$p(I, s, x)$ **Probability of intensity I at position x and scale s**

Weight → $W_D(s, x) = s \int \left| \frac{\partial}{\partial s} p(I, s, x) \right| dI$

Entropy → $H_D(s, x) = - \int p(I, s, x) \log_2 p(I, s, x) dI$

Scale value at which entropy function has a peak → $s_p = \left\{ s : \frac{\partial H_D(s, x)}{\partial s} = 0, \frac{\partial^2 H_D(s, x)}{\partial s^2} < 0 \right\}$

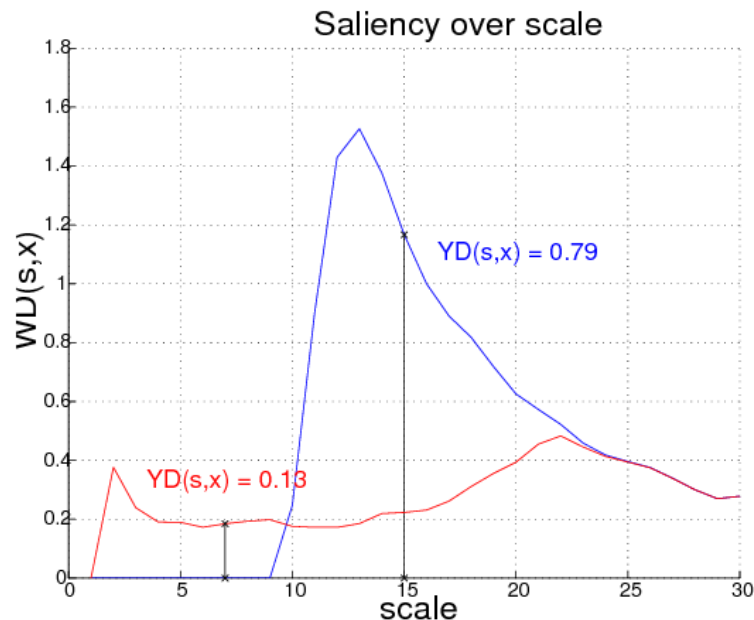
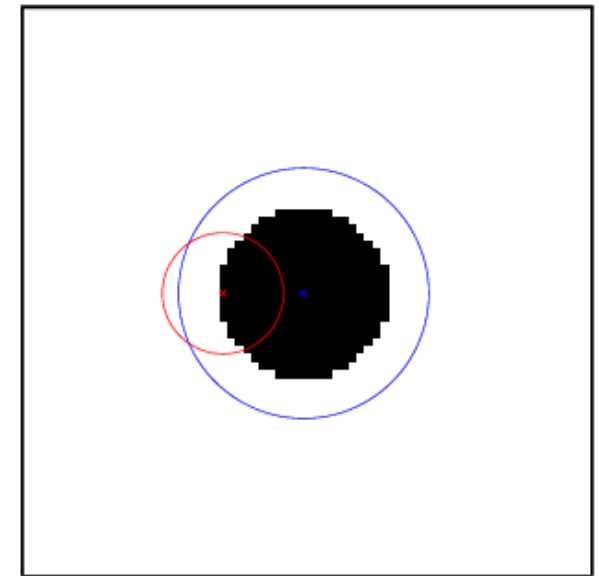
Salient Image Regions



Entropy as function of scale

Peaks in entropy

- Scales with equal number of black / white pixels
- Salient scales



Weight as function of scale

- Edge point has lower value than centre
- Normalized measure of scale localization
- Noisy image patch:
 - High entropy (spatially unpredictable)
 - Low saliency (similar statistics over scale)

Salient Image Regions

Algorithm

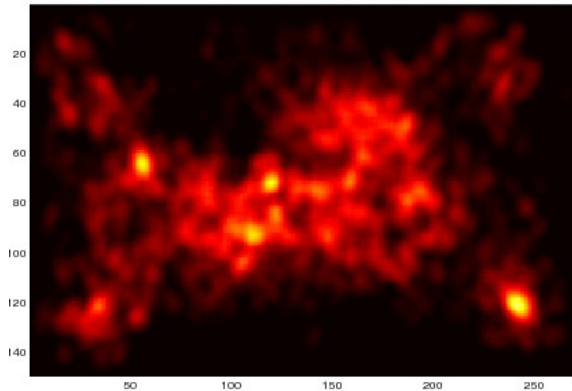
For each pixel location x :

1. For each scale s between s_{min} and s_{max} :
 - a) Measure the local descriptor values D within a window of scale s .
 - b) Estimate the local PDF (e.g. as histogram).
 - c) Calculate the local entropy H_D .
2. Select scales S for which the entropy is peaked (S may be empty).
3. Weight (W_D) the entropy values at S by the sum of absolute difference of the PDFs of the local descriptor around S .

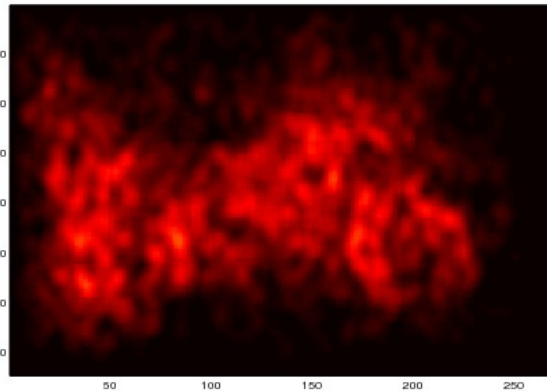
Salient Image Regions



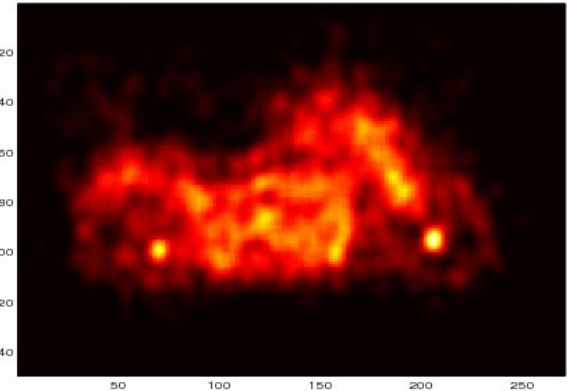
Database with side-views of motor bikes (with and without clutter)



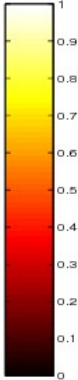
MultiScale Harris



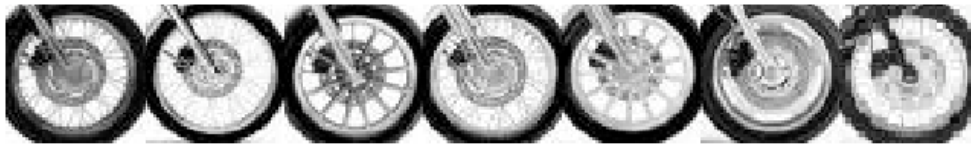
Difference-of-Gaussian



Saliency



Averaged detection of interest / salient points



Selected object parts

Saliency: Classical Approach

Most classical models

- are based on theoretical considerations about information theory
- use hand-crafted features / descriptors
- are not based on data
 - Data only used during testing / evaluation
 - Note: Exceptions are machine-learning based methods for predicting fixations and/or scan paths
- are often task independent

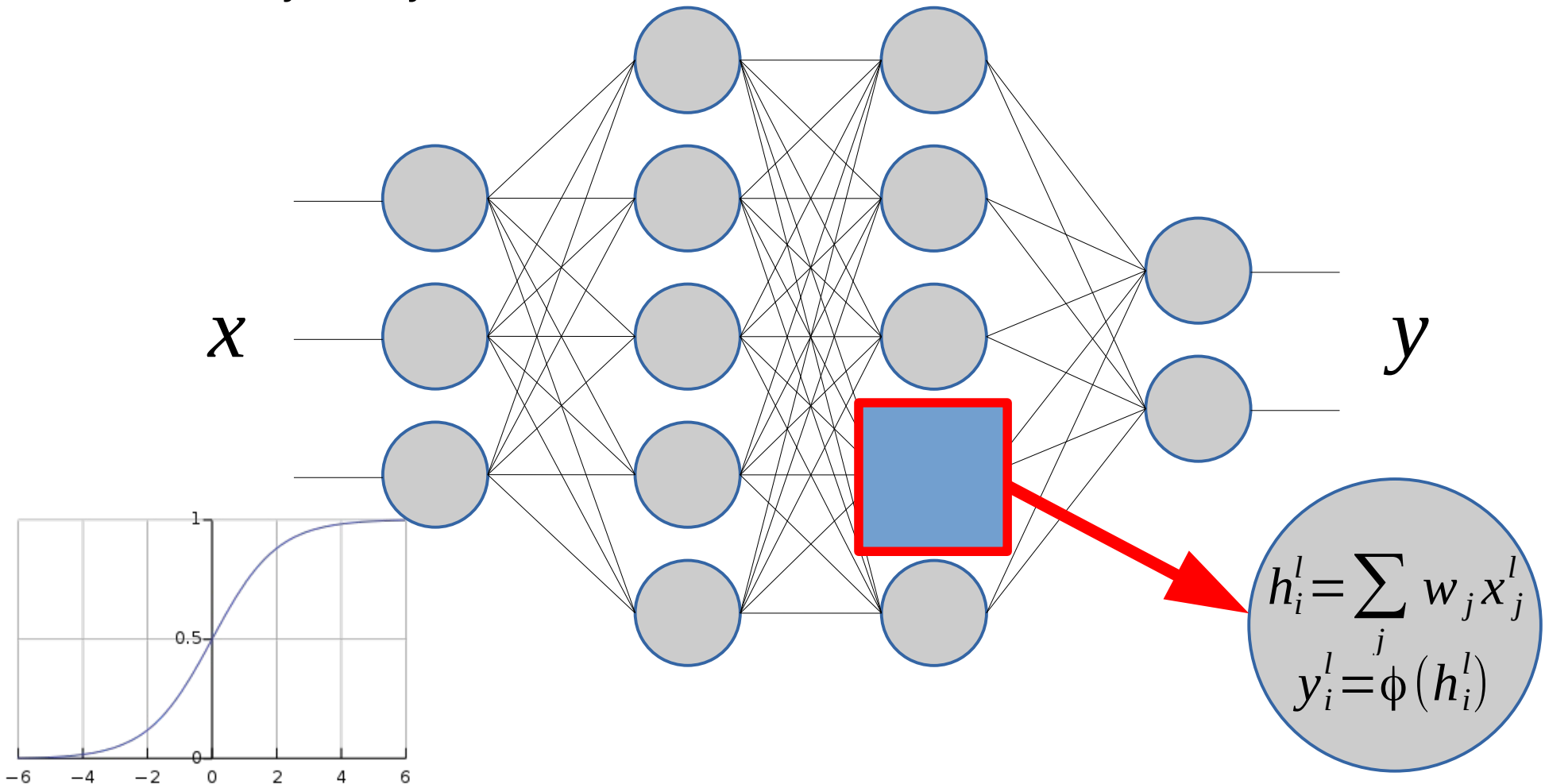
→ Modern alternative: Learning-based approaches

Deep learning

Saliency: Modern Approach

Reminder: MLP

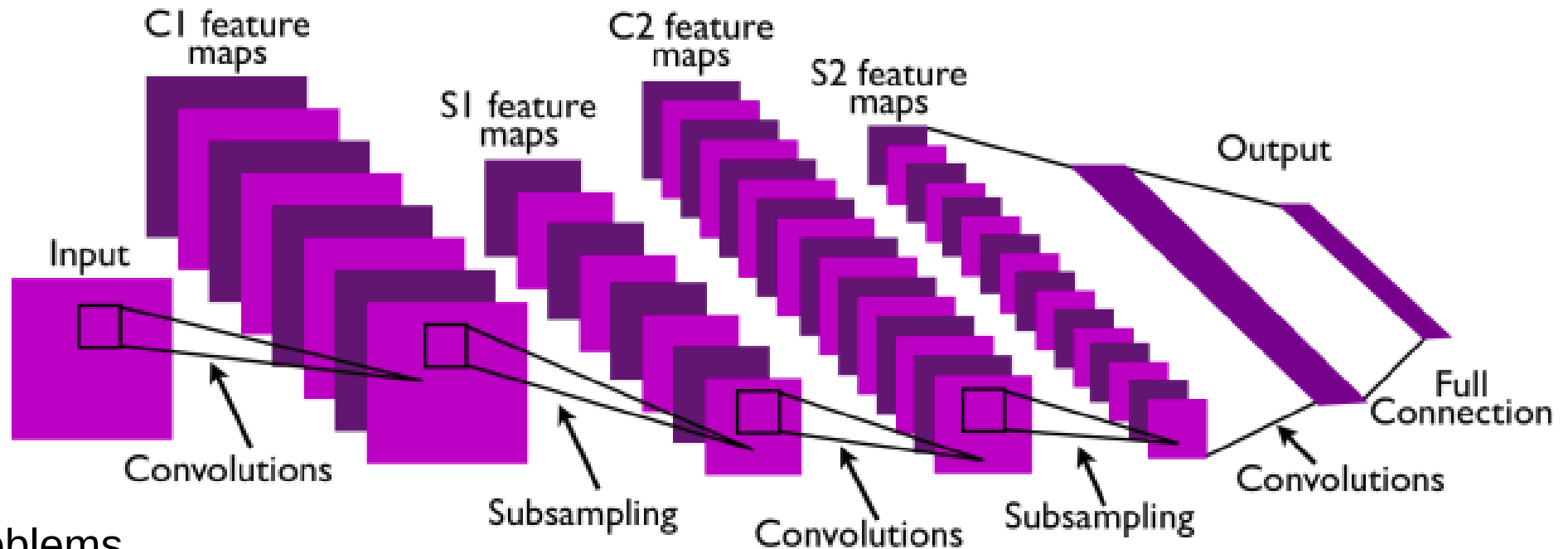
- Feed-forward artificial neural network
- Multiple layers of simple units (neurons)
- Each layer fully connected to the next one without lateral connections



Saliency: Modern Approach

Excursus: Convolutional (Neural) Network (ConvNet / CNN)

- Feed-forward artificial neural network
- Multiple layers of simple units (neurons)
- Convolutional layers, subsampling layers, fully-connected layers
- Trained in the same (similar) way as MLP: Backprop



Problems

- Many parameters → Needs A LOT of data
(- Training time → GPUs)

Visual Object Models

IV. Shape Models

Object Detection

Goal: Object categorization in real world scenes

GHT?

- Method should

Edges

Offset vectors

New R-Table

Voting Scheme

→ be based on appearance and shape.

→ be able to automatically adapt to new object classes.

→ be able to handle in-class variations of object examples.

→ During training: Given a collection of images of an object category

→ Learn an implicit shape model using local appearance

→ During application: Given an input image

→ Label and segment objects

Generalized Hough Transform

Training phase

- 1.1. Extract edge pixels from the template
- 1.2. Describe edge pixels based on gradient orientation
- 2.1. Group possible gradient orientations into intervals
- 2.2. Match edge pixels based on gradient orientation to groups in R-table
- 2.3. Store displacement vectors from edge position to reference point

$\theta = \frac{0 \cdot 2\pi}{A}$	→	$d_x^{1,1}, d_y^{(1,1)}$	...	d_x^{1,N_1}, d_y^{1,N_1}
$\theta = \frac{1 \cdot 2\pi}{A}$	→	$d_x^{2,1}, d_y^{(2,1)}$	...	d_x^{2,N_2}, d_y^{2,N_2}
$\theta = \frac{2 \cdot 2\pi}{A}$	→	$d_x^{3,1}, d_y^{(3,1)}$	...	d_x^{3,N_3}, d_y^{3,N_3}
$\vdots$	→		$\vdots$	
$\theta = \frac{(A-1) \cdot 2\pi}{A}$	→	$d_x^{A,1}, d_y^{(A,1)}$	...	d_x^{A,N_A}, d_y^{A,N_A}

Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**
- 2.1. Group possible **local appearances**
- 2.2. Match **object parts** based on **local appearance** to groups in **codebook**
- 2.3. Store displacement vectors from **object part** position to reference point

Implicit Shape Model: $ISM(C) = (I_C, P_{I,C})$

C : object category

I_C : class-specific alphabet (codebook)

$P_{I,C}$: spatial probability distribution

→ Specifies where each codebook entry may be found on the object.

→ Is defined independently for each codebook entry

- allows to combine object parts during recognition

GHT
Intervals of edge orientation
Offset vectors in R-table

Implicit Shape Model

Training phase

1.1. Extract **meaningful object parts** from the **training images**

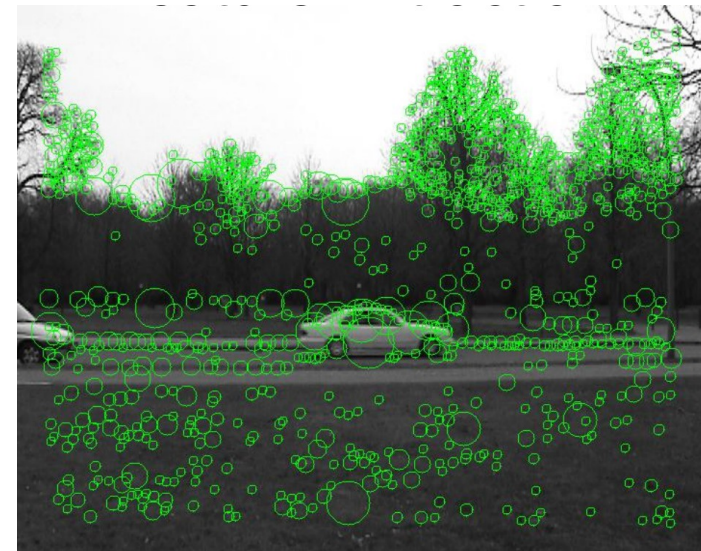
→ Feature extraction: Detection of salient points

- Interest points / Keypoints: Harris, Förstner, SIFT, etc. (see DIP)
- Salient Points: Kadir/Brady (see AIA - Saliency)

→ Scale σ (radius of salient point, σ of Gaussian function)

→ Defines a window size centred at the salient point

→ Such an image section is called a “patch”



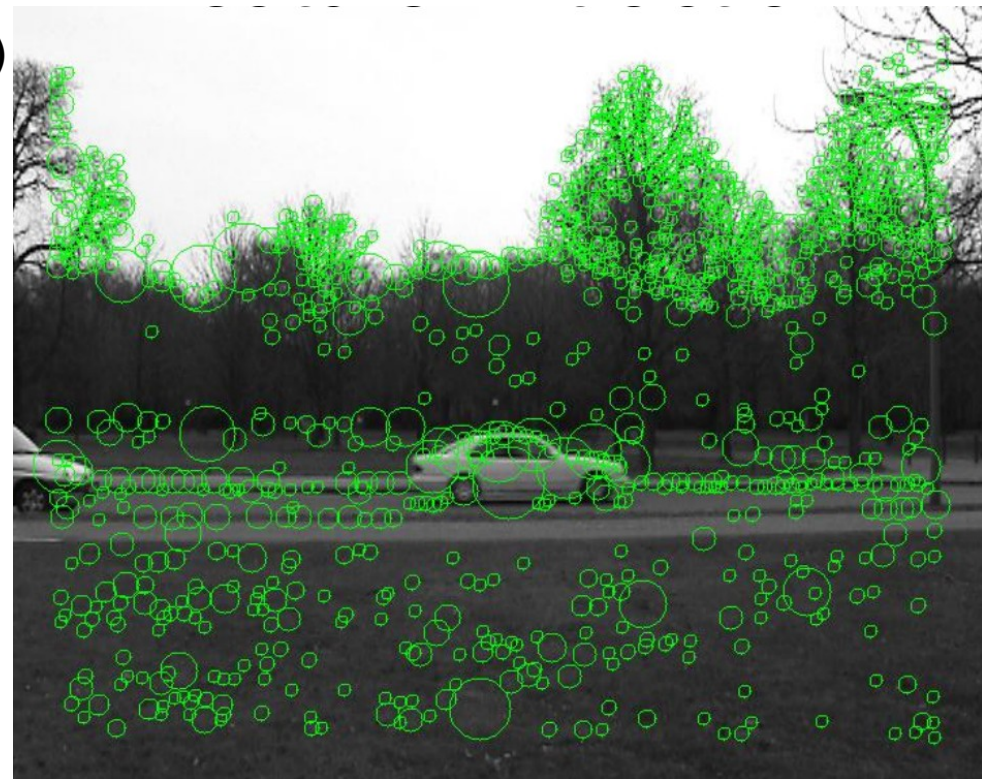
Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**

→ Feature description:

- local image intensity (patch)
- Keypoint descriptors (SIFT, SURF, etc.)



Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**
- 2.1. Group possible **local appearances**

→ Clustering:

- e.g. k-Means; here: iterative merging
 - Initially, each patch is treated as a separate cluster.
 - The two most similar clusters C_1 and C_2 are merged,
if the average similarity between their constituent patches
stays above a certain threshold t

Implicit Shape Model

→ The two most similar clusters C_1 and C_2 are merged,
if the average **similarity** between their constituent patches
stays above a certain threshold t

$$\text{similarity}(C_1, C_2) = \sum_{p \in C_1, q \in C_2} \frac{NGC(\mathbf{p}, \mathbf{q})}{|C_1||C_2|}$$

where $|C_i|$ is the number of patches in cluster C_i .
Normalized Greyscale Correlation:

$$NGC(\mathbf{p}, \mathbf{q}) = \frac{\sum_i (p_i - \bar{p})(q_i - \bar{q})}{\sqrt{\sum_i (p_i - \bar{p})^2 \sum_i (q_i - \bar{q})^2}}$$

where p_i is the grey value of a pixel in $\mathbf{p}$ and $\bar{p}$ the mean value of $\mathbf{p}$

Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**
- 2.1. Group possible **local appearances** (build **codebook**)

→ Clustering:

- e.g. k-Means; here: iterative merging
 - Initially, each patch is treated as a separate cluster.
 - The two most similar clusters C_1 and C_2 are merged,
if the average similarity between their constituent patches
stays above a certain threshold t
- Only patches are grouped which are visually similar.
- Clusters stay compact.

→ For each cluster, compute the cluster centre and store it in the codebook.

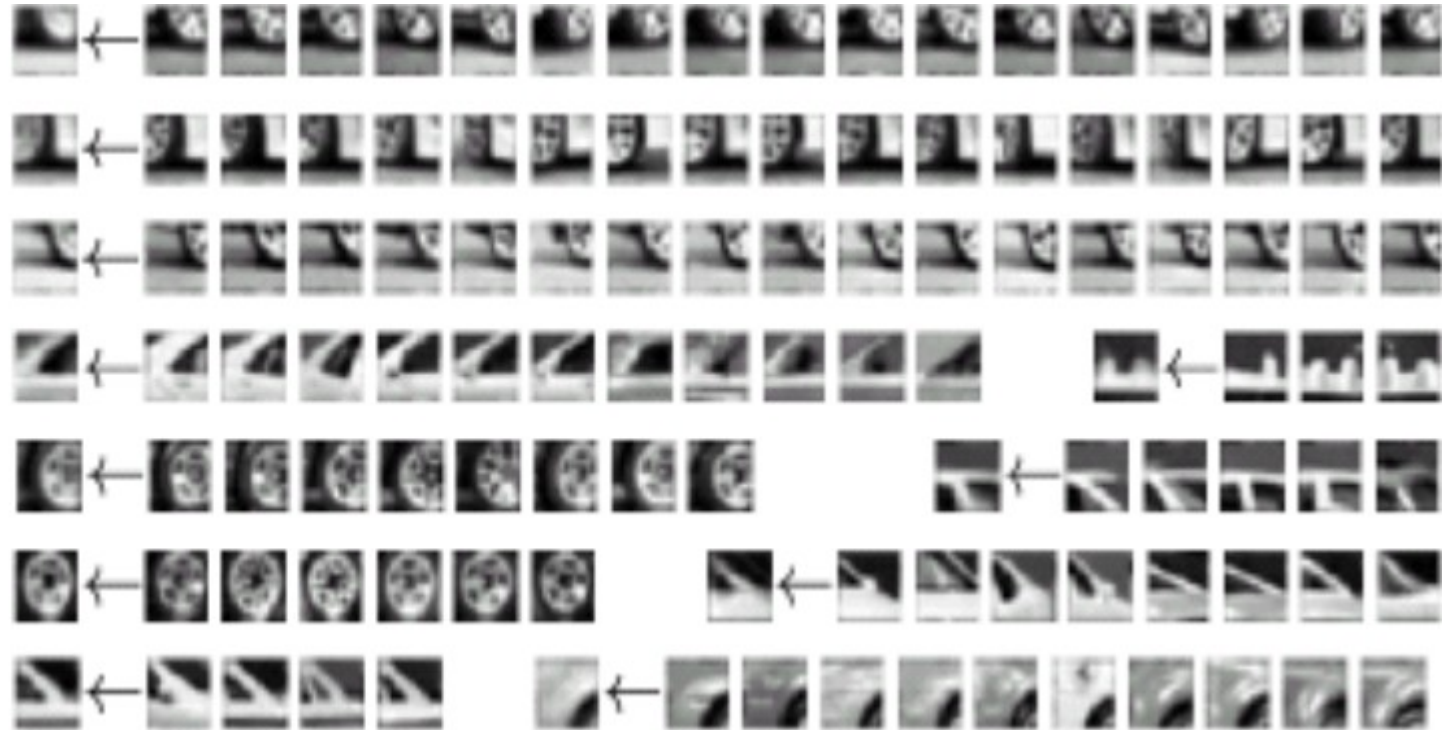
→ Do this for each training image.

Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**
- 2.1. Group possible **local appearances**

→ Clustering:



Implicit Shape Model

Training phase

- 1.1. Extract **meaningful object parts** from the **training images**
- 1.2. Describe **object parts** based on **local appearance (image content)**
- 2.1. Group possible **local appearances**
- 2.2. Match **object parts** based on **local appearance** to groups in **codebook**
- 2.3. Store displacement vectors from **object part** position to reference point

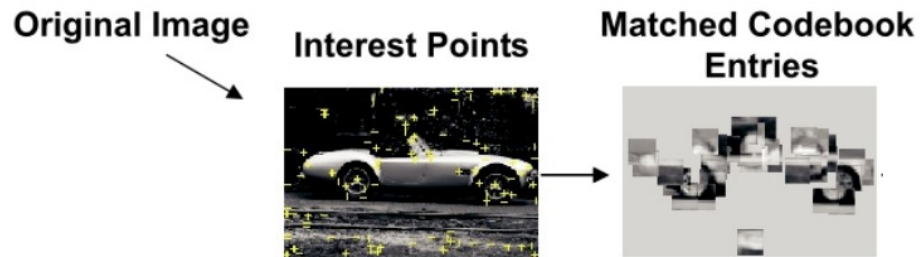
→ Codebook Entry Activation

- Iteration over all training images
 - Match codebook entries to image patches
 - Activate all entries whose similarity is above t
 - Assign offset vector from patch to object centre to the activated entry.
- During recognition, this information is used to perform the voting.

Implicit Shape Model

Prediction phase

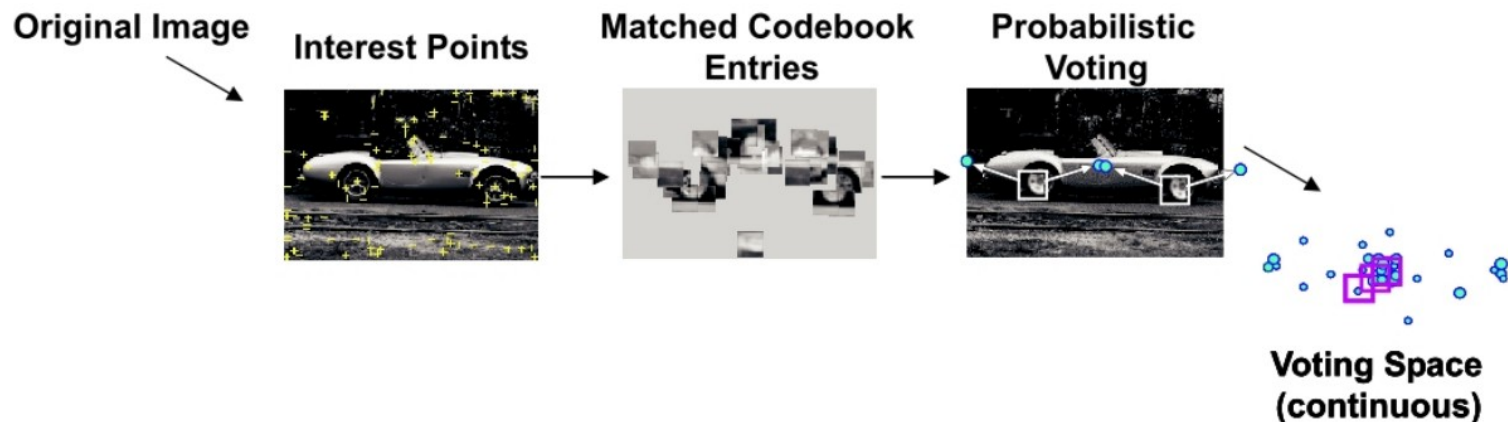
1. Extract salient points (=patches) from test image
 2. Match them to the codebook entries
- If similarity between patch and codebook entry is greater t
→ Activate codebook entry



Implicit Shape Model

Prediction phase

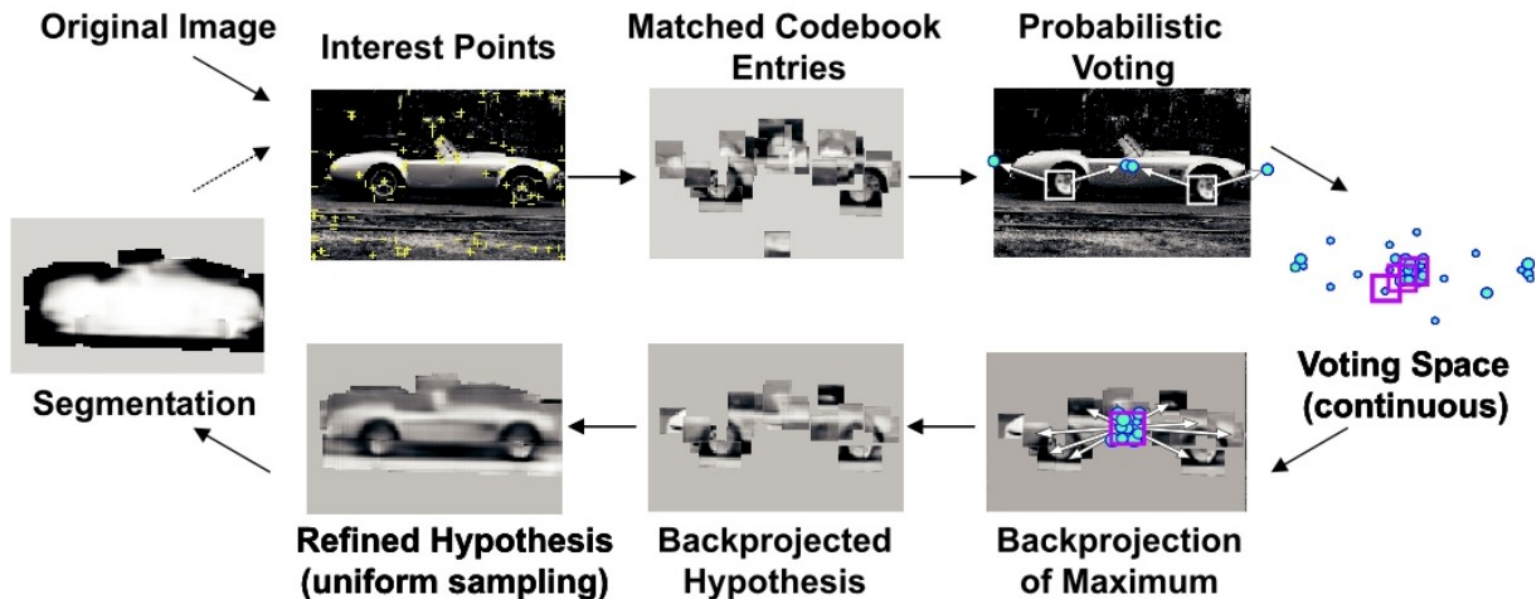
1. Extract salient points (=patches) from test image
2. Match them to the codebook entries
3. Generalized Hough voting:
 - Each activated entry casts votes for possible positions of the object centre.
 - A continuous voting space is used.
 - Use Mean-Shift Mode Estimation to search maxima (see DIP).
 - Alternative: non-parametric probability density estimation.



Implicit Shape Model

Prediction phase

1. Extract salient points (=patches) from test image
2. Match them to the codebook entries
3. Generalized Hough voting:
 - Each activated entry casts votes for possible positions of the object centre.
4. Backprojection
 - Project codebook entries from object centre into image space.
 - Results in a representation of the object including a certain border area.



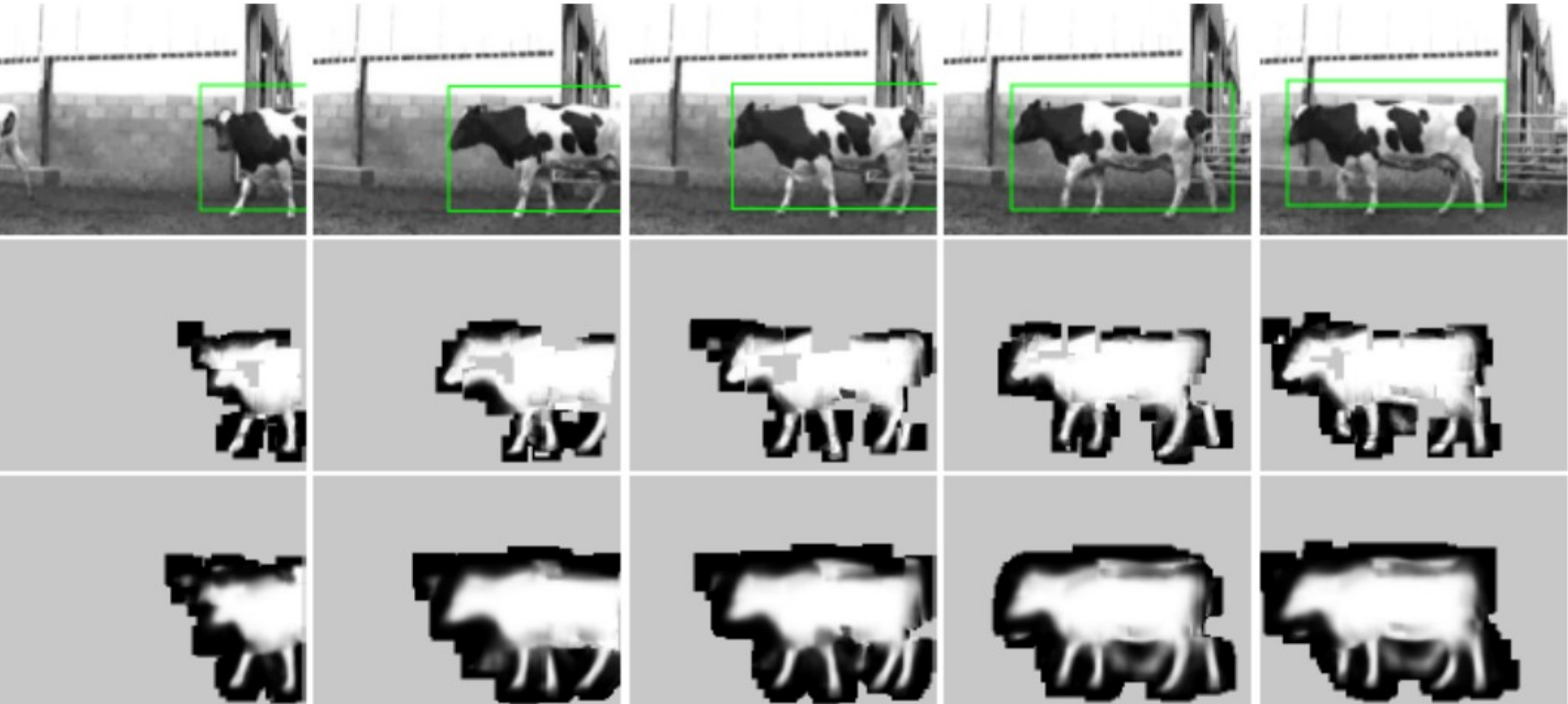
Implicit Shape Model

Segmentation



Implicit Shape Model

Probabilistic Formulation - Segmentation



Implicit Shape Model: Reference

- Leibe et al., Combined Object Categorization and Segmentation with an Implicit Shape Model, in ECCV'04 Workshop on Statistical Learning in Computer Vision, Prague, May 2004.
- B Leibe, A Leonardis, B Schiele:
Robust Object Detection with Interleaved Categorization and Segmentation,
International Journal of Computer Vision,
2008